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p-ADIC DIRAC EQUATIONS, CONTINUOUS-TIME QUANTUM WALKS, AND QUANTUM NETWORKS

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Abstract. We introduce a new class of p-adic Dirac equations, formulated in the standard axiomatic framework of quantum mechanics, in which the ordinary momentum operators are replaced by bounded non-local operators built from arbitrary integrable radial kernels. We diagonalize the resulting free Dirac Hamiltonian in momentum space, construct its plane-wave solutions, and determine its spectrum, which turns out to be pure point, with eigenvalues of infinite multiplicity; in the massless case we describe the zero-energy subspace explicitly. We identify the charge-conjugation symmetry of the model: it is implemented by an antiunitary operator whose square is -1, it exchanges the positive- and negative-energy subspaces of the free Hamiltonian, and in an external field it maps the charge e and vector potential $\boldsymbol{A}$ to $-e$ and $-\boldsymbol{A}$. We then discretize the free equation in two ways, both giving genuine continuous-time quantum walks rather than discrete-time, coined walks: a first construction on a countable graph, and a second, more explicit construction on a finite, tree-like graph. In both cases the discretization is exact: the walk is the restriction of the p-adic Dirac evolution to an invariant subspace, and its Hamiltonian is a Dirac-type combination of weighted graph Laplacians with ultrametric weights. We prove that the Born-rule transition probabilities, summed over the internal degrees of freedom and averaged over the initial internal state, form a doubly stochastic matrix at every instant; we also show that, as for any non-trivial continuous-time quantum walk, this family of matrices is not a Markov semigroup. Building on these results, we discuss how the construction can serve as the foundation of a quantum network with four-component spinor degrees of freedom, complementing earlier, non-relativistic p-adic quantum neural networks, and we prove global well-posedness of the associated nonlinear network equation under Lipschitz assumptions. We close with a discussion of the open mathematical and computational problems raised by this construction.

1. Introduction

Non-Archimedean, and specifically p-adic, models of physical space have been studied since the late 1980s as a way of exploring what quantum mechanics might look like if the Planck length is treated not as a mere order-of-magnitude estimate but as an actual discreteness scale built into the geometry of space itself, [1–6]. Replacing $\mathbb{R}$ with $\mathbb{Q}_p$, the field of p-adic numbers, has two immediate consequences: the resulting space is a totally disconnected, ultrametric, Cantor-like set rather than a continuum, and the Planck length can be identified with p^{-1} in a mathematically precise sense. Working within the Dirac–von

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Neumann axiomatic formulation of quantum mechanics, [7,8], on the Hilbert space $L^2(\mathbb{Q}_p^N)$, one obtains a rigorous, non-relativistic p-adic quantum mechanics, [9–16].

Recently, the author and his collaborators have shown that a large class of p-adic Schrödinger equations are scaling limits of continuous-time quantum walks (CTQWs), including CTQWs on graphs, [11–13]. This CTQW–Schrödinger correspondence was, in turn, used to build p-adic quantum neural networks (QNNs), whose neurons are organized in the hierarchical, tree-like structure inherited from $\mathbb{Z}_p$ (the unit ball in $\mathbb{Q}_p$) rather than on a Euclidean lattice, [17]. Furthermore, the author recently proposed a new collapse mechanism for wave-functions (using $(\mathbb{R}\times\mathbb{Q}_p)^3$ as model of the physical space, so that the model of the space-time is $\mathbb{R}\times(\mathbb{R}\times\mathbb{Q}_p)^3$) that provides a new explanation of the Wigner's friend paradox, [15, 16].

The construction of a p-adic version of the Dirac equation is very delicate, due to the relativistic nature of this equation and the fact that using a space-time (or configuration space) of type $\mathbb{R}\times\mathbb{Q}_p^N$ is not compatible with special relativity. In [10], the author introduced a p-adic Dirac equation, built from the Vladimirov pseudo-differential operator, which plays the role of the derivative on $\mathbb{Q}_p$, and showed that an isolated p-adic Dirac system does *not* satisfy Einstein causality: the discreteness of p-adic space allows correlations, and hence signals in a suitable sense, to propagate faster than the (finite, but p-dependent) analogue of the speed of light. The reader may consult [10, 11] for a further discussion on quantum non-locality and faster-than-light communication.

The p-adic Dirac equations are not relativistic: the space-time $\mathbb{R}\times\mathbb{Q}_p^3$ carries no analogue of the Lorentz group, and the equations are not invariant under any such group. They do, however, retain the algebraic and spectral structure of the Dirac equation. They are first-order equations for four-component spinors, whose free Hamiltonian is built from the Dirac matrices; they admit plane-wave solutions with the dispersion relation $E^2 = |\mathbf{j}(k)|^2 + m^2$, where $\mathbf{j}(k) \in \mathbb{R}^3$ is the symbol of the operators that replace the momentum (Section 4.3); their spectrum splits into positive- and negative-energy branches, which are exchanged by a charge-conjugation symmetry; and a Foldy–Wouthuysen transformation decouples them into a pair of square-root Klein–Gordon equations. It is in this sense that we call them Dirac equations. A topologically protected extension of the equation of [10], modeled on the Jackiw–Rebbi soliton, was introduced more recently in [14].

Until now, these two threads—the p-adic Dirac equation on one side, the p-adic CTQW/QNN formalism on the other—have remained disjoint: no discretization of a p-adic Dirac equation into a continuous-time quantum walk had been constructed, and consequently no p-adic quantum network with spinor-like internal degrees of freedom was available. In the ordinary (Archimedean) literature, which we review in Section 2, the overwhelming majority of the works relating quantum walks to the Dirac equation use *discrete-time*, coined quantum walks, going back to Feynman's relativistic checkerboard and made rigorous by Bialynicki-Birula, Meyer, and Succi and Benzi, with the Dirac equation appearing as a continuum *limit* of the walk, [18–21]. Continuous-time constructions carrying a spinor-like structure also exist—for instance the CTQWs with internal degrees of freedom on periodic lattices used by Childs and Goldstone for spatial search, whose Hamiltonian is a lattice

discretization of the Dirac operator [22], the topological Dirac operator on networks and simplicial complexes introduced by Bianconi [23], CTQWs on bipartite graphs [24], and tight-binding Hamiltonians on the honeycomb lattice [25,26]—but none of them is formulated over a hierarchical, ultrametric graph.

The present paper fills this gap in the non-Archimedean setting. We introduce a different, complementary class of p-adic Dirac equations, in which the momentum operator along each spatial direction x_l (the analogue of $-\mathrm{i}\partial/\partial x_l$) is replaced by a non-local difference operator $\boldsymbol{J}_{x_l}$ built from an *arbitrary* integrable radial kernel $J_l(|y|_p)$ (consequently, the free Hamiltonian is $\boldsymbol{\alpha}\cdot\nabla+\beta m$, without the factor $-\mathrm{i}$ of the textbook form $-\mathrm{i}\boldsymbol{\alpha}\cdot\nabla_{\mathbb{R}}+\beta m$, which is already absorbed into the operators $\boldsymbol{J}_{x_l}$; see Remark 4.1), rather than from the Vladimirov pseudo-differential operator used in [10]. Unlike the Vladimirov operator, which is unbounded on $L^2(\mathbb{Q}_p)$, the operator $\boldsymbol{J}_{x_l}$ is bounded for every integrable kernel J_l; consequently, the equation of [10] is *not* a particular case of the family introduced here, nor is the present family a generalization of it in the strict sense—the two constructions are built from operators of genuinely different analytic type, neither contained in the other. The boundedness of $\boldsymbol{J}_{x_l}$ is, however, precisely what makes this new family well suited to discretization: it yields a large, explicit class of p-adic Dirac equations, parametrized by the choice of J_1, J_2, J_3, each member of which can be discretized directly into a continuous-time quantum walk.

We carry out, for this new class, the programme of analysis that is classical for the standard Dirac equation, [29–31]: we diagonalize the free Hamiltonian $\boldsymbol{H}_0$ in the Fourier representation, construct its plane-wave solutions, determine its spectrum and show that it is contained in the spectrum of the standard free Dirac operator, exhibit the Foldy–Wouthuysen-type transformation that decouples $\boldsymbol{H}_0$ into a pair of square-root Klein–Gordon operators, and study charge conjugation. Two features distinguish the p-adic model from the standard one. First, the spectrum of $\boldsymbol{H}_0$ is pure point, every eigenvalue has infinite multiplicity, and in the massless case $\boldsymbol{H}_0$ may have an infinite-dimensional kernel (Theorem 6.1 and Proposition 6.2). Second, since the operators $\boldsymbol{J}_{x_l}$ commute with complex conjugation (whereas $-\mathrm{i}\partial/\partial x_l$ anticommutes with it), the charge-conjugation operator is $\mathcal{C}\Psi = \alpha_2\overline{\Psi}$, which satisfies $\mathcal{C}^2 = -1$, and in the presence of an electromagnetic field it reverses both the charge and the vector potential (Lemma 7.1).

We then construct two discretizations of the free equation into CTQWs: a first construction on the countable graph $\widetilde{G}_n^3 = \mathbb{Q}_p^3/p^n\mathbb{Z}_p^3$ (Section 8), and a second, more explicit construction on the finite graph $G_n^3 = \mathbb{Z}_p^3/p^n\mathbb{Z}_p^3$ (Section 9). The key observation is that the space of square-integrable functions that are constant on the balls of radius p^{-n} is invariant under $\boldsymbol{H}_0$, and that, on this space, each $\boldsymbol{J}_{x_l}$ acts as a weighted graph Laplacian whose weights are determined by the p-adic tree. Consequently, the restriction of the p-adic Dirac evolution to this space is exactly a CTQW, in the sense of Farhi and Gutmann [39], with a $\mathbb{C}^4$ internal degree of freedom at each vertex; no continuum limit is involved. For these walks we prove that the Born-rule transition probabilities, summed over the four spinor components, define a stochastic matrix for every t, and that, after averaging over the initial internal state,

this matrix is doubly stochastic (Theorem 9.2). As for any CTQW, the resulting family of stochastic matrices is not a Markov semigroup (Proposition 9.1).

Beyond their intrinsic mathematical interest, these results have a direct implication for quantum networks, which we develop in Section 10. The constructions of Sections 8 and 9 suggest that the free p-adic Dirac CTQW can serve as the substrate of a quantum network in the sense of [17], extended here from scalar, non-relativistic internal degrees of freedom to four-component spinor degrees of freedom. We write the corresponding nonlinear network equation precisely and prove its global well-posedness under Lipschitz assumptions (Proposition 10.1). We close the paper, in Section 11, with a discussion of the mathematical open problems that this construction raises.

In a forthcoming paper, we plan to extend the numerical methods introduced in [13] and [17] to the case of the p-adic Dirac equations and the associated QNNs.

The paper is organized as follows. Section 2 reviews the state of the art on the relation between quantum walks and the Dirac equation, both in the Archimedean and in the non-Archimedean setting. Section 3 collects the notation and basic facts on p-adic analysis used throughout the paper (a more detailed review is given in the Appendix). Section 4 introduces the new class of p-adic Dirac equations. Section 5 constructs their plane-wave solutions. Section 6 diagonalizes the free Hamiltonian, determines its spectrum, describes the positive-, negative- and zero-energy subspaces, and establishes the Foldy–Wouthuysen decoupling. Section 7 studies charge conjugation. Sections 8 and 9 construct the two CTQW discretizations of the free equation. Section 10 discusses the resulting quantum-network construction, and Section 11 concludes with a discussion of open problems. An Appendix collects the background material on p-adic analysis needed in the body of the paper.

2. Related work

The idea that the Dirac equation is, at bottom, a quantum walk on a lattice goes back to Feynman's relativistic checkerboard model for the $1+1$ dimensional propagator, [32]. This intuition was made mathematically precise, independently, by Bialynicki-Birula, [18], Meyer, [19], and Succi and Benzi, [20], who showed that a homogeneous (translation-invariant, time-independent), strictly causal discrete-time quantum walk (DTQW) on $\mathbb{Z}$, with a two-dimensional coin space attached to each site, converges, as the lattice spacing $\varepsilon \to 0$, to the exact solution of the free Dirac equation. Arrighi, Nesme, and Forets extended this construction to $2+1$ and $3+1$ dimensions and proved consistency, stability, and convergence of the corresponding DTQWs to the higher-dimensional Dirac equation, [21]. This body of work established what is now the standard paradigm: *the Dirac equation is the continuum limit of a discrete-time, coined quantum walk*, with the coin space playing the role of the spinor degrees of freedom.

A second, closely related line of research uses this correspondence to locate genuinely relativistic phenomena—Zitterbewegung and the Klein paradox—inside the purely quantum-informational dynamics of the DTQW. Strauch obtained closed-form Bessel-function solutions of the one-dimensional DTQW and showed that, in one limit, they reduce to the

wave-packet solutions of the free Dirac equation, exhibiting relativistic spreading, while in the opposite limit they reduce to the solutions of the ordinary continuous-time quantum walk (CTQW), [33, 34]. Kurzyński compared the one-dimensional DTQW directly with the discretized Dirac equation and showed that both Zitterbewegung and Klein's paradox are already present in the walk, [35]. Chandrashekar, Banerjee, and Srikanth pushed this comparison further, relating the coupled and decoupled forms of the DTQW to the free Dirac and Klein–Gordon equations, respectively, and identifying the walk's coin as the analogue of the spinor degree of freedom, [36]. These results were subsequently realized experimentally on a trapped-ion platform: Gerritsma, Kirchmair, Zähringer, Solano, Blatt, and Roos implemented the $1+1$ dimensional Dirac equation by coupling a single qubit to the vibrational mode of a trapped ion and observed Zitterbewegung directly, [37], and the same group later used the same platform to simulate the Klein paradox, [38].

A parallel line of research studies *continuous*-time quantum walks. The CTQW was introduced by Farhi and Gutmann as the direct quantization of a classical continuous-time random walk on a graph, [39]: its dynamics is governed by a Schrödinger equation $\mathrm{i}\partial_t\psi = H\psi$, with H the graph adjacency matrix or the graph Laplacian. The framework was subsequently developed and surveyed extensively, e.g., in [40–42]. In [11], the author established that the Farhi–Gutmann CTQWs are a particular case of a much larger family of CTQWs constructed using techniques of p-adic analysis, as we discuss below. For this reason the CTQW is naturally the continuous-time counterpart of the *Schrödinger*, rather than the Dirac, equation, and the literature connecting CTQWs directly to Dirac dynamics is thinner than that for its discrete-time counterpart.

Several constructions do, however, inject a genuine spinor-like structure into continuous-time dynamics on graphs. Childs and Goldstone [22] showed that the continuous-time quantum-walk search algorithm on low-dimensional periodic lattices can be substantially improved by enlarging the Hilbert space with internal (spin) degrees of freedom; their walk Hamiltonian is a lattice discretization of the massless Dirac operator, and the Dirac equation governs its long-wavelength behaviour. Bianconi [23] introduced the topological Dirac operator of a network (and, more generally, of a simplicial complex), which acts on "topological spinors" formed by node and link signals, and the associated topological Dirac equation, which generates a continuous-time unitary dynamics on an arbitrary graph. Tödtli, Laner, Semenov, Paoli, Blattner, and Kunegis studied CTQWs on directed bipartite graphs and showed that probability transport between the two node partitions—which can be read as the two components of a spinor—can be tuned continuously, and even completely suppressed, by a single phase parameter, [24]. Finally, continuous-time tight-binding Hamiltonians on the honeycomb lattice, introduced to model electron transport in graphite and graphene [25, 26], provide another continuous-time realization of Dirac dynamics, since their low-energy dispersion around the Dirac points is that of the $2+1$ dimensional massless Dirac equation; the corresponding *discrete-time* quantum walks on triangular and honeycomb lattices, and their continuum limits, were constructed by Jay, Debbasch, and Wang [27].

Di Molfetta and Debbasch classified which continuum limits are attainable from families of coined walks and showed that, when a continuum limit exists, it is described either by a Dirac-like equation or by a pair of decoupled Klein–Gordon equations, [43]; the same authors, together with Brachet, extended this program to curved space-time, showing that a suitably chosen family of DTQWs converges to massless Dirac fermions propagating on a given background metric, [44]. Arnault and Debbasch developed gauge-covariant discrete-time walks [28].

None of the constructions reviewed above is formulated over a non-Archimedean field, or over a hierarchical (ultrametric) graph. On the p-adic side, the author showed that free p-adic Schrödinger equations arise as the scaling limit of continuous-time quantum Markov chains on the p-adic tree $\mathbb{Z}_p$, and that discretizing these equations produces genuine CTQWs on finite graphs, [11]; this correspondence was developed further, together with Chacón-Cortés, through detailed comparisons and numerical simulations of the resulting discretizations, [13], and, together with Mayes, in the setting of p-adic infinite potential wells, [12]. On the equation side, a p-adic Dirac equation was introduced in [10], where it was shown to violate Einstein causality as a consequence of the discreteness of p-adic space; a topologically protected version of this equation, modeled on the Jackiw–Rebbi soliton, was introduced more recently in [14]. These two threads—p-adic CTQWs on one side, the p-adic Dirac equation on the other—have, until now, remained disjoint: no discretization of a p-adic Dirac equation into a continuous-time quantum walk has been constructed. The present paper fills this gap. In contrast with the Archimedean constructions above, where the Dirac equation appears as a long-wavelength or continuum limit of the walk, here the walk is the *exact* restriction of the p-adic Dirac evolution to an invariant subspace (Theorems 8.1 and 9.1). We note that a companion construction, restricted to the p-adic Schrödinger case and aimed at quantum neural network architectures, was developed in [17].

3. Some mathematical preliminaries

From now on, we use p to denote a fixed prime number. Any non-zero p-adic number x has a unique expansion of the form

$$x = x_{-k}p^{-k} + x_{-k+1}p^{-k+1} + \ldots + x_0 + x_1 p + \ldots, \tag{3.1}$$

with $x_{-k} \neq 0$, where k is an integer, and the x_js are numbers from the set $\{0, 1, \ldots, p-1\}$. The set of all possible sequences of the form (3.1) constitutes the field of p-adic numbers $\mathbb{Q}_p$. There are natural field operations, sum and multiplication, on series of form (3.1). There is also a norm in $\mathbb{Q}_p$ defined as $|x|_p = p^{-ord(x)}$, where $ord(x) = -k$, for a nonzero p-adic number x. By definition $ord(0) = \infty$. The field of p-adic numbers with the distance induced by $|\cdot|_p$ is a complete ultrametric space. The ultrametric property refers to the fact that $|x-y|_p \leq \max\{|x-z|_p, |z-y|_p\}$ for any x, y, z in $\mathbb{Q}_p$. The p-adic integers are sequences of form (3.1) with $-k \geq 0$. All these sequences constitute the unit ball $\mathbb{Z}_p$. The unit ball is an infinite rooted tree with fractal structure. As a topological space $\mathbb{Q}_p$ is homeomorphic to a Cantor-like subset of the real line, see, e.g., [6, 45].

We extend the p-adic norm to $\mathbb{Q}_p^N$ by taking

$$\|x\|_p := \max_{1\leq i\leq N} |x_i|_p, \qquad \text{for } x=(x_1,\ldots,x_N)\in \mathbb{Q}_p^N.$$

We define $ord(x)=\min_{1\leq i\leq N}\{ord(x_i)\}$, then $\|x\|_p = p^{-ord(x)}$. The metric space $(\mathbb{Q}_p^N, \|\cdot\|_p)$ is a complete ultrametric space.

A function $\varphi : \mathbb{Q}_p^N \to \mathbb{C}$ is called locally constant, if for any $a\in\mathbb{Q}_p^N$, there is an integer $l=l(a)$, such that $\varphi(a+x)=\varphi(a)$ for any $\|x\|_p \leq p^l$. If φ is locally constant and has compact support, we say that φ is a test function. We denote by $\mathcal{D}(\mathbb{Q}_p^N)$ the complex vector space of test functions. There is a natural integration theory so that $\int_{\mathbb{Q}_p^N}\varphi(x)d^Nx$ gives a well-defined complex number. The measure d^Nx is the Haar measure of $\mathbb{Q}_p^N$, normalized by $\int_{\mathbb{Z}_p^N} d^Nx = 1$. In the Appendix, we give a quick review of the basic aspects of the p-adic analysis required here.

By p-adic quantum mechanics (QM), we mean QM in the sense of the Dirac–von Neumann formulation on the Hilbert space

$$L^2(\mathbb{Q}_p^N) := L^2(\mathbb{Q}_p^N, d^Nx) = \left\{ f : \mathbb{Q}_p^N \to \mathbb{C}; \|f\|_2 = \left(\int_{\mathbb{Q}_p^N} |f(x)|^2 d^Nx \right)^{\frac{1}{2}} < \infty \right\}.$$

Given $f,g\in L^2(\mathbb{Q}_p^N)$, we set $\langle f,g\rangle = \int_{\mathbb{Q}_p^N} f(x)\overline{g(x)}d^Nx$, where the bar denotes the complex conjugate.

4. A new class of p-adic Dirac equations

4.1. **Abstract Dirac equations.** We denote the Pauli matrices as

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}, \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

where $\mathrm{i}=\sqrt{-1}\in\mathbb{C}$, and the 4×4 Dirac matrices as

$$\beta = \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{bmatrix}, \qquad \alpha_l = \begin{bmatrix} \mathbf{0} & \sigma_l \\ \sigma_l & \mathbf{0} \end{bmatrix}, \quad l=1,2,3,$$

where $\mathbf{1}$ denotes the 2×2 identity matrix, and $\mathbf{0}$ denotes the 2×2 zero matrix. These matrices are Hermitian and satisfy the anticommutation relations

$$\alpha_l\alpha_k + \alpha_k\alpha_l = 2\delta_{lk}I_4, \qquad \alpha_l\beta+\beta\alpha_l = 0, \qquad \beta^2 = I_4, \tag{4.1}$$

where I_4 is the 4×4 identity matrix; note that $I_4=\mathrm{diag}(\mathbf{1},\mathbf{1})$ (block-diagonal), where $\mathbf{1}$ is the 2×2 identity matrix (identity operators on other spaces are denoted by Id). Note that α_1, α_3, β are real, while $\overline{\alpha_2}=-\alpha_2$.

We set

$$\nabla := \begin{bmatrix} \boldsymbol{J}_{x_1} \\ \boldsymbol{J}_{x_2} \\ \boldsymbol{J}_{x_3} \end{bmatrix},$$

where $\boldsymbol{J}_{x_1}$, $\boldsymbol{J}_{x_2}$, $\boldsymbol{J}_{x_3}$ are pairwise commuting self-adjoint operators on $L^2(\mathbb{X}^3)$, which play the role of an abstract version of the momentum operators $-\mathrm{i}\frac{\partial}{\partial x_l}$. The spatial variables x_1, x_2, x_3 run in an abstract space $\mathbb{X}$, which can be specialized to $\mathbb{R}$. We use the standard notation

$$\boldsymbol{\alpha}\cdot\nabla := \sum_{l=1}^{3}\alpha_l \boldsymbol{J}_{x_l}, \qquad \boldsymbol{\sigma}\cdot\nabla := \sum_{l=1}^{3}\sigma_l \boldsymbol{J}_{x_l}.$$

An abstract Dirac equation is an equation of the form

$$\mathrm{i}\frac{\partial}{\partial t}\Psi(t,x) = (\boldsymbol{\alpha}\cdot\nabla + \beta m)\,\Psi(t,x), \qquad t\in\mathbb{R},\ x\in\mathbb{X}^3, \tag{4.2}$$

where $m\geq 0$ (the mass), and $\Psi^T(t,x) = [\Psi_1(t,x)\ \Psi_2(t,x)\ \Psi_3(t,x)\ \Psi_4(t,x)]\in\mathbb{C}^4$. If $\mathbb{X}=\mathbb{R}$ and $\boldsymbol{J}_{x_l} = -\mathrm{i}\frac{\partial}{\partial x_l}$, then (4.2) is the standard free Dirac equation (in units with $\hbar = c = 1$).

We define the free Hamiltonian as the operator

$$\boldsymbol{H}_0 = \boldsymbol{\alpha}\cdot\nabla + \beta m = \begin{bmatrix} m\mathbf{1} & \boldsymbol{\sigma}\cdot\nabla \\ \boldsymbol{\sigma}\cdot\nabla & -m\mathbf{1}\end{bmatrix}.$$

Then, our abstract version of the Dirac equation can be rewritten as

$$\mathrm{i}\frac{\partial}{\partial t}\Psi(t,x) = \boldsymbol{H}_0\Psi(t,x), \qquad t\in\mathbb{R},\ x\in\mathbb{X}^3. \tag{4.3}$$

Remark 4.1 (On the absence of the factor $-\mathrm{i}$ in $\boldsymbol{H}_0$)**.** *The reader used to the standard Dirac equation should note that our free Hamiltonian is written as* $\boldsymbol{H}_0 = \boldsymbol{\alpha}\cdot\nabla + \beta m$, without *the factor* $-\mathrm{i}$ *that appears in the textbook form* $\boldsymbol{H}_0^{Arch} = -\mathrm{i}\boldsymbol{\alpha}\cdot\nabla_{\mathbb{R}} + \beta m$, *where* $\nabla_{\mathbb{R}} = (\partial_{x_1},\partial_{x_2},\partial_{x_3})$*; see, e.g., [29, Chapter 1]. This is not an omission: in this article the components of* ∇ *are the analogues of the self-adjoint momentum operators* $-\mathrm{i}\partial/\partial x_l$*, not of the skew-adjoint operators* $\partial/\partial x_l$*, and the factor* $-\mathrm{i}$ *is already absorbed into them. Thus, for* $\mathbb{X}=\mathbb{R}$*, one recovers* $\boldsymbol{H}_0^{Arch}$ *by taking* $\boldsymbol{J}_{x_l} = -\mathrm{i}\partial/\partial x_l$.

This convention is forced in the p-adic setting. The operators $\boldsymbol{J}_{x_l}$ *introduced in Section 4.3 have real, even kernels; they are self-adjoint (Section 4.2), like* $-\mathrm{i}\partial/\partial x_l$*. If one inserted an additional factor* $-\mathrm{i}$*, i.e., used* $-\mathrm{i}\boldsymbol{\alpha}\cdot\nabla+\beta m$*, the resulting operator would not be self-adjoint, since* $(-\mathrm{i}\alpha_l\boldsymbol{J}_{x_l})^* = \mathrm{i}\alpha_l\boldsymbol{J}_{x_l}$*. Indeed, its symbol* $\widetilde{h}(\xi) = -\mathrm{i}\boldsymbol{\alpha}\cdot\mathbf{j}(\xi)+\beta m$ *(notation of Section 4.3) satisfies, by (4.1),* $\widetilde{h}(\xi)^2 = \left(m^2-|\mathbf{j}(\xi)|^2\right)I_4$*, so its eigenvalues* $\pm\sqrt{m^2-|\mathbf{j}(\xi)|^2}$ *are purely imaginary whenever* $|\mathbf{j}(\xi)|>m$ *(for instance, at every* ξ *with* $\mathbf{j}(\xi)\neq 0$ *when* $m=0$*), and the corresponding evolution* $e^{-\mathrm{i}t\widetilde{h}(\xi)}$ *grows exponentially in* t*. The associated dynamics would not be unitary, and the probabilistic interpretation of Sections 8 and 9 would be lost.*

Equivalently, if one prefers to keep the textbook form of the Hamiltonian, one may introduce the p-adic analogues of the derivatives, $\mathcal{D}_{x_l} := \mathrm{i}\boldsymbol{J}_{x_l}$*, which are skew-adjoint, like* $\partial/\partial x_l$*; then* $-\mathrm{i}\sum_l \alpha_l\mathcal{D}_{x_l} + \beta m = \boldsymbol{\alpha}\cdot\nabla+\beta m = \boldsymbol{H}_0$*. The two descriptions define the same operator; we use the one in terms of the self-adjoint operators* $\boldsymbol{J}_{x_l}$ *throughout. Finally, we note that, since the* $\boldsymbol{J}_{x_l}$ *commute with complex conjugation (Remark 4.2) while* $-\mathrm{i}\partial/\partial x_l$ *anticommutes with it, the charge-conjugation operator of the p-adic model differs from the standard one; see Section 7.*

4.2. **The $\boldsymbol{J}$ operator.** We take $\mathbb{X} = \mathbb{Q}_p$, the field of p-adic numbers. In the Appendix, we give a brief review of the essential ideas and results of p-adic analysis required in this paper.

We fix a function $J : [0, \infty) \to \mathbb{R}$ such that $y \mapsto J(|y|_p)$ is an integrable function on $\mathbb{Q}_p$, and set

$$\|J\|_1 := \int_{\mathbb{Q}_p} \Big| J(|y|_p) \Big| \, dy < \infty, \qquad \widehat{J}(0) := \int_{\mathbb{Q}_p} J(|y|_p) dy \in \mathbb{R}.$$

Here dy denotes the normalized Haar measure on $\mathbb{Q}_p$; see the Appendix. Notice that $\|J\|_1$ is the L^1-norm of $J(|y|_p)$, while $\widehat{J}(0)$ is the signed integral of $J(|y|_p)$; no sign condition is imposed on J, so $\widehat{J}(0)$ may be negative or zero, but always $\Big|\widehat{J}(0)\Big| \leq \|J\|_1$.

Now, for $\rho \in [1, \infty]$, we define the operator

$$\begin{array}{rccl} \boldsymbol{J}: & L^\rho(\mathbb{Q}_p) & \to & L^\rho(\mathbb{Q}_p) \\ & \phi(x) & \mapsto & \displaystyle\int_{\mathbb{Q}_p} J(|x-y|_p) \left\{\phi(y) - \phi(x)\right\} dy, \end{array}$$

where the integral is understood as $\boldsymbol{J}\phi(x) = (J(|\cdot|_p) * \phi)(x) - \widehat{J}(0)\phi(x)$; both terms are well defined for almost every x. By Young's inequality, $\Big\| J(|\cdot|_p) * \phi \Big\|_\rho \leq \|J\|_1 \, \|\phi\|_\rho$, and hence

$$\|\boldsymbol{J}\phi\|_\rho \leq \|J\|_1 \, \|\phi\|_\rho + \Big|\widehat{J}(0)\Big| \, \|\phi\|_\rho \leq 2 \, \|J\|_1 \, \|\phi\|_\rho \,,$$

for $\rho \in [1, \infty]$. Therefore $\boldsymbol{J}$ is a linear bounded operator from $L^\rho(\mathbb{Q}_p)$ into itself for $\rho \in [1, \infty]$. For instance, the kernel $J(|y|_p) = -\Omega(|y|_p)$, the negative of the characteristic function of $\mathbb{Z}_p$, is admissible; in this case $\widehat{J}(0) = -1$, $\|J\|_1 = 1$, and the above bound reads $\|\boldsymbol{J}\phi\|_\rho \leq 2 \, \|\phi\|_\rho$. Most of the time, we will take $\rho = 2$.

In the case $\rho = 2$, the operator $\boldsymbol{J}$ is pseudo-differential. Indeed, the Fourier transform of $J(|x|_p)$,

$$\mathcal{F}_{x \to \xi}(J(|x|_p)) = \int_{\mathbb{Q}_p} J(|x|_p) \chi_p(\xi x) dx = \widehat{J}(|\xi|_p),$$

is a continuous, real-valued, radial function, which tends to zero as $|\xi|_p \to \infty$ (Riemann–Lebesgue lemma), and whose value at $\xi = 0$ is $\widehat{J}(0)$. It is real-valued because $J(|-x|_p) = J(|x|_p)$, and radial because the Haar measure and $|\cdot|_p$ are invariant under $x \mapsto ax$ for $|a|_p = 1$. We set

$$\mathfrak{j}(\xi) := \widehat{J}(|\xi|_p) - \widehat{J}(0), \qquad \xi \in \mathbb{Q}_p. \tag{4.4}$$

The function $\mathfrak{j}$ is real-valued, radial, continuous, $\mathfrak{j}(0) = 0$, and $|\mathfrak{j}(\xi)| \leq 2 \, \|J\|_1$. Furthermore, for $\phi \in L^2(\mathbb{Q}_p)$,

$$\boldsymbol{J}\phi(x) = \mathcal{F}^{-1}_{\xi \to x} \Big\{ \mathfrak{j}(\xi) \widehat{\phi}(\xi) \Big\} \in L^2(\mathbb{Q}_p),$$

where $\widehat{\phi}(\xi)$ denotes the Fourier transform of $\phi(x)$; in particular, the operator norm of $\boldsymbol{J}$ on $L^2(\mathbb{Q}_p)$ is $\sup_\xi |\mathfrak{j}(\xi)| \leq 2 \, \|J\|_1$. A particularly important formula is

$$\boldsymbol{J}\chi_p(kx) = \mathfrak{j}(k)\chi_p(kx) = \Big\{ \widehat{J}(|k|_p) - \widehat{J}(0) \Big\} \chi_p(kx), \qquad \text{for } k, \ x \in \mathbb{Q}_p. \tag{4.5}$$

Here, χ_p denotes the standard additive character of $(\mathbb{Q}_p,+)$, which is the p-adic analogue of the complex exponential function; see the Appendix. The function $x\mapsto \chi_p(kx)$ does not belong to $L^2(\mathbb{Q}_p)$, but it is bounded, so the integral defining $\boldsymbol{J}\chi_p(kx)$ converges absolutely; indeed, changing variables $y=x-z$,

$$\boldsymbol{J}\chi_p(kx)=\chi_p(kx)\int_{\mathbb{Q}_p} J(|z|_p)\left\{\chi_p(-kz)-1\right\}dz=\chi_p(kx)\left\{\widehat{J}(|k|_p)-\widehat{J}(0)\right\}.$$

The operator $\boldsymbol{J}$ is self-adjoint in $L^2(\mathbb{Q}_p)$: since the Fourier transform preserves the inner product and $\mathfrak{j}$ is real-valued,

$$\begin{aligned}\langle\psi,\boldsymbol{J}\phi\rangle&=\int_{\mathbb{Q}_p}\psi(x)\overline{\boldsymbol{J}\phi(x)}dx=\int_{\mathbb{Q}_p}\widehat{\psi}(\xi)\overline{\mathcal{F}_{x\to\xi}\{\boldsymbol{J}\phi(x)\}}d\xi=\int_{\mathbb{Q}_p}\widehat{\psi}(\xi)\,\mathfrak{j}(\xi)\,\overline{\widehat{\phi}(\xi)}d\xi\\&=\int_{\mathbb{Q}_p}\boldsymbol{J}\psi(x)\overline{\phi(x)}dx=\langle\boldsymbol{J}\psi,\phi\rangle.\end{aligned}$$

Along the paper, we use the symbol $\langle\cdot,\cdot\rangle$ to denote the inner product of $L^2(\mathbb{Q}_p^N)$. The corresponding norm is denoted as $\|\cdot\|_2$.

Example 4.1. *Take $J(|y|_p)=\Omega(|y|_p)$, the characteristic function of $\mathbb{Z}_p$. Then $\widehat{J}(|\xi|_p)=\Omega(|\xi|_p)$, $\widehat{J}(0)=1$, $\mathfrak{j}(\xi)=-1_{\mathbb{Q}_p\smallsetminus\mathbb{Z}_p}(\xi)$, and $\boldsymbol{J}\phi(x)=\int_{x+\mathbb{Z}_p}\phi(y)dy-\phi(x)$, i.e., $\boldsymbol{J}$ is minus the orthogonal projection onto the functions whose Fourier transform is supported outside $\mathbb{Z}_p$. If J is supported in the ball $B_L=p^{-L}\mathbb{Z}_p$, then $\chi_p(\xi y)=1$ for $|\xi|_p\leq p^{-L}$ and $y\in B_L$, hence $\widehat{J}(|\xi|_p)=\widehat{J}(0)$ and $\mathfrak{j}(\xi)=0$ for $|\xi|_p\leq p^{-L}$.*

Remark 4.2. *On $\mathbb{R}$, the momentum operator $-\mathrm{i}\frac{d}{dx}$ is a Fourier multiplier with a real,* odd *symbol. The operator $\boldsymbol{J}$ is a Fourier multiplier with a real,* even *(indeed radial) symbol $\mathfrak{j}$. Since $\mathbb{Q}_p$ is not an ordered field, radial symbols are the natural ones in the p-adic setting: for instance, the Vladimirov operator, used in [10], has symbol $|\xi|_p$, which is the analogue of the symbol of $|d/dx|$. In the abstract equation (4.2), the operators $\boldsymbol{J}_{x_l}$ enter only through the identity $(\boldsymbol{\alpha}\cdot\nabla)^2=\sum_l\boldsymbol{J}_{x_l}^2$, which follows from (4.1) and the commutativity of the $\boldsymbol{J}_{x_l}$; this is what produces the Dirac-type dispersion relation of Section 5. The parity of the symbol, on the other hand, determines the behaviour under complex conjugation: since the kernel J is real, $\overline{\boldsymbol{J}\phi}=\boldsymbol{J}\overline{\phi}$, whereas $\overline{-\mathrm{i}\phi'}=-(-\mathrm{i}\overline{\phi}')$. This difference has consequences for charge conjugation; see Section 7.*

4.3. **A new type of p-adic Dirac equations.** We now fix three functions $J_l:[0,\infty)\to\mathbb{R}$, $l=1,2,3$, such that $J_l(|y|_p)$ is an integrable function on $\mathbb{Q}_p$, and set

$$\boldsymbol{J}_{x_l}\phi(x):=\int_{\mathbb{Q}_p}J_l(|x_l-y|_p)\left\{\phi(x^{(l)}(y))-\phi(x)\right\}dy,\qquad l=1,2,3,\tag{4.6}$$

for $\phi\in L^2(\mathbb{Q}_p^3)$, where $x=(x_1,x_2,x_3)$ and $x^{(l)}(y)$ denotes the point obtained from x by replacing its l-th coordinate by y. In other words, $\boldsymbol{J}_{x_l}$ is the operator $\boldsymbol{J}$ of the previous subsection, built from the kernel J_l, acting on the variable x_l. We denote by $\mathfrak{j}_l(\xi_l)=\widehat{J}_l(|\xi_l|_p)-\widehat{J}_l(0)$ the corresponding symbols, and set $\nabla^T:=[\boldsymbol{J}_{x_1}\ \boldsymbol{J}_{x_2}\ \boldsymbol{J}_{x_3}]$.

Notation. For $\xi = (\xi_1, \xi_2, \xi_3) \in \mathbb{Q}_p^3$, we set

$$\mathbf{j}(\xi) := (\mathfrak{j}_1(\xi_1), \mathfrak{j}_2(\xi_2), \mathfrak{j}_3(\xi_3)) \in \mathbb{R}^3, \qquad |\mathbf{j}(\xi)|^2 := \sum_{l=1}^{3} \mathfrak{j}_l(\xi_l)^2,$$

$$\boldsymbol{\alpha} \cdot \mathbf{j}(\xi) := \sum_{l=1}^{3} \alpha_l \, \mathfrak{j}_l(\xi_l), \qquad \boldsymbol{\sigma} \cdot \mathbf{j}(\xi) := \sum_{l=1}^{3} \sigma_l \, \mathfrak{j}_l(\xi_l),$$

and

$$\lambda(\xi) := \sqrt{|\mathbf{j}(\xi)|^2 + m^2}. \tag{4.7}$$

Notice that $\mathbf{j}(\xi)$ and $\lambda(\xi)$ depend only on $(|\xi_1|_p, |\xi_2|_p, |\xi_3|_p)$. By (4.1), $(\boldsymbol{\alpha} \cdot \mathbf{j}(\xi))^2 = |\mathbf{j}(\xi)|^2 I_4$ and $(\boldsymbol{\sigma} \cdot \mathbf{j}(\xi))^2 = |\mathbf{j}(\xi)|^2 \mathbf{1}$. We also recall that $k \cdot x = \sum_{i=1}^{3} k_i x_i \in \mathbb{Q}_p$, for $k, x \in \mathbb{Q}_p^3$.

In the next three sections, we will show that

$$\mathrm{i} \frac{\partial}{\partial t} \Psi(t,x) = \boldsymbol{H}_0 \Psi(t,x), \qquad t \in \mathbb{R}, \ x \in \mathbb{Q}_p^3, \tag{4.8}$$

with $\boldsymbol{H}_0 = \boldsymbol{\alpha} \cdot \nabla + \beta m$ and ∇ built from the operators (4.6), is a new family of p-adic Dirac equations.

5. Plane waves

Definition 5.1. *Fix $k \in \mathbb{Q}_p^3$ such that $\lambda(k) + m > 0$ (this holds for every k if $m > 0$; see Remark 5.1 for the case $\lambda(k) + m = 0$). We define the spinors*

$$w^{(1)}(k) = \begin{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \dfrac{\boldsymbol{\sigma} \cdot \mathbf{j}(k)}{\lambda(k) + m} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix}, \qquad w^{(2)}(k) = \begin{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \dfrac{\boldsymbol{\sigma} \cdot \mathbf{j}(k)}{\lambda(k) + m} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix}, \tag{5.1}$$

$$w^{(3)}(k) = \begin{bmatrix} \dfrac{-\boldsymbol{\sigma} \cdot \mathbf{j}(k)}{\lambda(k) + m} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix}, \qquad w^{(4)}(k) = \begin{bmatrix} \dfrac{-\boldsymbol{\sigma} \cdot \mathbf{j}(k)}{\lambda(k) + m} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix}, \tag{5.2}$$

and the energies $E_1(k) = E_2(k) = \lambda(k)$, $E_3(k) = E_4(k) = -\lambda(k)$. By a plane wave, we mean a function of the form

$$\Psi_s(t,x) = e^{-\mathrm{i} E_s(k) t} \chi_p(k \cdot x) \, w^{(s)}(k), \qquad x \in \mathbb{Q}_p^3, \ t \in \mathbb{R}, \ s = 1,2,3,4. \tag{5.3}$$

Thus Ψ_1, Ψ_2 are proportional to $e^{-\mathrm{i}\lambda(k)t}$ (positive energy), while Ψ_3, Ψ_4 are proportional to $e^{+\mathrm{i}\lambda(k)t}$ (negative energy).

The operator $\boldsymbol{H}_0$ is defined on bounded functions by the integral formula (4.6); in particular, $\boldsymbol{H}_0 \Psi_s(t,\cdot)$ makes sense although $\Psi_s(t,\cdot) \notin L^2(\mathbb{Q}_p^3) \otimes \mathbb{C}^4$. As in the classical case, plane waves are generalized eigenfunctions; superpositions $\int \widehat{\psi}(k) \Psi_s(t,x) \, d^3k$ with $\widehat{\psi} \in L^2(\mathbb{Q}_p^3)$ give genuine solutions; see Section 6.

Proposition 5.1. *The plane waves (5.3) are solutions of the p-adic Dirac equation (4.8). More precisely, setting*

$$h(k) := \boldsymbol{\alpha}\cdot\mathbf{j}(k)+\beta m=\begin{bmatrix} m\mathbf{1} & \boldsymbol{\sigma}\cdot\mathbf{j}(k)\\ \boldsymbol{\sigma}\cdot\mathbf{j}(k) & -m\mathbf{1}\end{bmatrix},$$

we have $h(k)w^{(s)}(k)=E_s(k)w^{(s)}(k)$ for $s=1,2,3,4$. Moreover, the vectors $w^{(s)}(k)$ are pairwise orthogonal in $\mathbb{C}^4$, and $\left|w^{(s)}(k)\right|^2=2\lambda(k)/(\lambda(k)+m)$.

Proof. By (4.5), applied in each variable x_l, $\boldsymbol{J}_{x_l}\chi_p(k\cdot x)=\mathfrak{j}_l(k_l)\chi_p(k\cdot x)$, since $\chi_p(k\cdot x)=\prod_l\chi_p(k_lx_l)$. Hence $\boldsymbol{H}_0\left(\chi_p(k\cdot x)w\right)=\chi_p(k\cdot x)h(k)w$ for every $w\in\mathbb{C}^4$, and, since $\mathrm{i}\frac{\partial}{\partial t}e^{-\mathrm{i}Et}=Ee^{-\mathrm{i}Et}$, the function (5.3) solves (4.8) if and only if

$$E_s(k)w^{(s)}(k)=h(k)w^{(s)}(k). \tag{5.4}$$

Explicitly, writing $\mathfrak{j}_l=\mathfrak{j}_l(k_l)$, (5.4) is the linear system $(h(k)-EI_4)w=0$, where

$$h(k)-EI_4=\begin{bmatrix} m-E & 0 & \mathfrak{j}_3 & \mathfrak{j}_1-\mathrm{i}\mathfrak{j}_2\\ 0 & m-E & \mathfrak{j}_1+\mathrm{i}\mathfrak{j}_2 & -\mathfrak{j}_3\\ \mathfrak{j}_3 & \mathfrak{j}_1-\mathrm{i}\mathfrak{j}_2 & -m-E & 0\\ \mathfrak{j}_1+\mathrm{i}\mathfrak{j}_2 & -\mathfrak{j}_3 & 0 & -m-E\end{bmatrix},$$

whose determinant is $(m^2+\left|\mathbf{j}(k)\right|^2-E^2)^2$; thus non-trivial solutions exist if and only if $E=\pm\lambda(k)$. We verify (5.4) directly. Write $S:=\boldsymbol{\sigma}\cdot\mathbf{j}(k)$, so that $S^2=\left|\mathbf{j}(k)\right|^2\mathbf{1}=(\lambda^2-m^2)\mathbf{1}$, with $\lambda=\lambda(k)$. For $\eta\in\mathbb{C}^2$ and $w=\left[\eta,\ \frac{S\eta}{\lambda+m}\right]^T$,

$$h(k)w=\begin{bmatrix} m\eta+\frac{S^2\eta}{\lambda+m}\\ S\eta-\frac{mS\eta}{\lambda+m}\end{bmatrix}=\begin{bmatrix}(m+\lambda-m)\eta\\ \frac{\lambda}{\lambda+m}S\eta\end{bmatrix}=\lambda w,$$

which gives the cases $s=1,2$. For $w=\left[-\frac{S\eta}{\lambda+m},\ \eta\right]^T$,

$$h(k)w=\begin{bmatrix} -\frac{mS\eta}{\lambda+m}+S\eta\\ -\frac{S^2\eta}{\lambda+m}-m\eta\end{bmatrix}=\begin{bmatrix}\frac{\lambda}{\lambda+m}S\eta\\ -\lambda\eta\end{bmatrix}=-\lambda w,$$

which gives the cases $s=3,4$. Finally, S is Hermitian, so $\left|w^{(s)}\right|^2=1+\frac{|S\eta|^2}{(\lambda+m)^2}=1+\frac{\lambda-m}{\lambda+m}=\frac{2\lambda}{\lambda+m}$ for $|\eta|=1$; $w^{(1)}\perp w^{(2)}$ and $w^{(3)}\perp w^{(4)}$ by a direct computation, and the eigenvectors of the Hermitian matrix $h(k)$ for the distinct eigenvalues $\pm\lambda$ are orthogonal (if $\lambda=0$ then $\lambda+m=0$, which is excluded). The computation is the same as in the classical case; see, e.g., [30, 31]. □

Remark 5.1. *The denominators in (5.1)–(5.2) vanish if and only if $\lambda(k)+m=0$, i.e., if and only if $m=0$ and $\mathbf{j}(k)=0$. In this case $h(k)=0$, and every function $\chi_p(k\cdot x)w$, $w\in\mathbb{C}^4$, is a time-independent (zero-energy) solution of (4.8). This happens, for instance, for $k=0$, and, by Example 4.1, for every k in the ball $\{|k_l|_p\leq p^{-L_l},\ l=1,2,3\}$ if each J_l is supported in $p^{-L_l}\mathbb{Z}_p$. These zero-energy modes have no analogue for the massive standard Dirac equation, and they are discussed in Section 6.4. If $m>0$, then $\lambda(k)+m\geq 2m>0$ for every k, and no singularity occurs. The normalized spinors are $\sqrt{(\lambda(k)+m)/(2\lambda(k))}\,w^{(s)}(k)$.*

6. The free p-adic Dirac operator and its spectrum

6.1. **Some function spaces.** This section uses a notation similar to the one used in [29, Chapter 1] to compare the standard and the p-adic Dirac operators quickly. Furthermore, we use several results and calculations in [29, Chapter 1]. We set

$$\mathfrak{H} := L^2(\mathbb{Q}_p^3) \otimes \mathbb{C}^4,$$

and identify $\mathfrak{H}$, via the isomorphism of complex Hilbert spaces $\phi \otimes e_j \mapsto (0, \dots, \phi, \dots, 0)$ (ϕ in the j-th place), with the orthogonal direct sum $\bigoplus_{j=1}^4 L^2(\mathbb{Q}_p^3) = L^2(\mathbb{Q}_p^3)^4$. Thus we identify the elements of $\mathfrak{H}$ with column vectors of the form $\psi(x) = [\psi_1(x), \dots, \psi_4(x)]^T$, $x \in \mathbb{Q}_p^3$. The inner product is given by

$$(\psi, \phi) = \int_{\mathbb{Q}_p^3} \sum_{l=1}^4 \psi_l(x)\overline{\phi_l(x)}\, d^3x = \sum_{l=1}^4 \langle \psi_l, \phi_l \rangle,$$

and the norm by $\|\psi\| = \sqrt{\sum_{l=1}^4 \|\psi_l\|_2^2}$. Given $\psi \in \mathfrak{H}$ with integrable components, its Fourier transform is defined as $(\mathcal{F}\psi)(\xi) = \widehat{\psi}(\xi) = [\widehat{\psi}_1(\xi), \dots, \widehat{\psi}_4(\xi)]^T$, where

$$(\mathcal{F}\psi_i)(\xi) = \widehat{\psi}_i(\xi) = \int_{\mathbb{Q}_p^3} \chi_p(\xi \cdot x)\psi_i(x)\, d^3x, \qquad \xi \in \mathbb{Q}_p^3,$$

for $i = 1, 2, 3, 4$. The Fourier transform extends to a uniquely defined unitary operator (denoted as $\mathcal{F}$) on the Hilbert space $\mathfrak{H}$.

Proposition 6.1. *The mapping $\psi \mapsto \boldsymbol{H}_0\psi$ is a bounded, self-adjoint, linear operator on $\mathfrak{H}$, with $\|\boldsymbol{H}_0\| \leq M := \left(m^2 + 4\sum_{l=1}^3 \|J_l\|_1^2\right)^{1/2}$. Furthermore, $\{e^{-\mathrm{i}t\boldsymbol{H}_0}\}_{t\in\mathbb{R}}$ is a norm-continuous group of unitary operators on $\mathfrak{H}$, and, for every $\psi_0 \in \mathfrak{H}$, the Cauchy problem*

$$\begin{cases} \mathrm{i}\frac{\partial}{\partial t}\Psi(t,x) = \boldsymbol{H}_0\Psi(t,x), & t \in \mathbb{R},\ x \in \mathbb{Q}_p^3, \\ \Psi(0,x) = \psi_0(x), \end{cases}$$

has a unique solution $\Psi \in C^1(\mathbb{R}, \mathfrak{H})$, given by $\Psi(t,x) = e^{-\mathrm{i}t\boldsymbol{H}_0}\psi_0(x)$. In particular, the restriction $\{e^{-\mathrm{i}t\boldsymbol{H}_0}\}_{t\geq 0}$, which describes the evolution forward in time, is a unitary semigroup.

Proof. Each $\boldsymbol{J}_{x_l}$ is a bounded self-adjoint operator on $L^2(\mathbb{Q}_p^3)$, being unitarily equivalent, via $\mathcal{F}$, to the multiplication by the bounded real function $\mathfrak{j}_l(\xi_l)$; and each α_l, and β, is a Hermitian matrix. Hence each term $\alpha_l\boldsymbol{J}_{x_l}$ (i.e., $\boldsymbol{J}_{x_l} \otimes \alpha_l$) and βm is bounded and self-adjoint on $\mathfrak{H}$, and so is $\boldsymbol{H}_0$. The bound on $\|\boldsymbol{H}_0\|$ follows from the description of $\boldsymbol{H}_0$ in Fourier space given in Section 6.2: $\|\boldsymbol{H}_0\| = \sup_\xi \lambda(\xi)$ and $|\mathfrak{j}_l| \leq 2\|J_l\|_1$. Since $\boldsymbol{H}_0$ is bounded, $e^{-\mathrm{i}t\boldsymbol{H}_0} = \sum_{k\geq 0}(-\mathrm{i}t\boldsymbol{H}_0)^k/k!$ converges in norm, it is unitary because $\boldsymbol{H}_0$ is self-adjoint, $t \mapsto e^{-\mathrm{i}t\boldsymbol{H}_0}$ is a norm-continuous group, and $\Psi(t) = e^{-\mathrm{i}t\boldsymbol{H}_0}\psi_0$ is a C^1 solution. If $\Phi \in C^1(\mathbb{R}, \mathfrak{H})$ is another solution, then $\frac{d}{dt}e^{\mathrm{i}t\boldsymbol{H}_0}\Phi(t) = 0$, so $\Phi(t) = e^{-\mathrm{i}t\boldsymbol{H}_0}\psi_0$. □

6.2. **The free Dirac operator in the Fourier space.** The results presented in this section are analogs of the results for the standard Dirac operator. In particular, the calculations used here are the same as the ones given in [29, Section 1.4.1]. The Hamiltonian $\boldsymbol{H}_0$ is a matrix pseudo-differential operator on $\mathfrak{H}$, which is transformed via $\mathcal{F}$ into a matrix multiplication operator:

$$(\boldsymbol{H}_0\phi)(x) = \mathcal{F}^{-1}_{\xi\to x}\left(h(\xi)\mathcal{F}_{x\to\xi}\phi\right), \qquad \text{for } \phi\in\mathfrak{H}, \tag{6.1}$$

where

$$h(\xi) := \boldsymbol{\alpha}\cdot\mathbf{j}(\xi) + \beta m = \begin{bmatrix} m\mathbf{1} & \boldsymbol{\sigma}\cdot\mathbf{j}(\xi) \\ \boldsymbol{\sigma}\cdot\mathbf{j}(\xi) & -m\mathbf{1}\end{bmatrix}.$$

The matrix $h(\xi)$ is a 4×4 Hermitian matrix satisfying $h(\xi)^2 = \lambda(\xi)^2 I_4$, by (4.1). It has the eigenvalues $\lambda_1(\xi)=\lambda_2(\xi)=-\lambda_3(\xi)=-\lambda_4(\xi)=\lambda(\xi)$, with $\lambda(\xi)$ as in (4.7).

We define the 4×4 matrix

$$u(\xi) := \frac{(m+\lambda(\xi))I_4 + \beta\,\boldsymbol{\alpha}\cdot\mathbf{j}(\xi)}{\sqrt{2\lambda(\xi)(m+\lambda(\xi))}} \quad \text{if } \lambda(\xi)>0, \qquad u(\xi) := I_4 \quad \text{if } \lambda(\xi)=0. \tag{6.2}$$

Notice that $\lambda(\xi)=0$ if and only if $m=0$ and $\mathbf{j}(\xi)=0$; in this case $h(\xi)=0$. When $\mathbf{j}(\xi)\neq 0$, $u(\xi)$ can be rewritten as

$$u(\xi) = a_+(\xi)I_4 + a_-(\xi)\beta\frac{\boldsymbol{\alpha}\cdot\mathbf{j}(\xi)}{|\mathbf{j}(\xi)|}, \qquad a_\pm(\xi) = \frac{1}{\sqrt{2}}\sqrt{1\pm\frac{m}{\lambda(\xi)}},$$

while $u(\xi)=I_4$ when $\mathbf{j}(\xi)=0$ (because then $\lambda(\xi)=m$).

Lemma 6.1. *For every $\xi\in\mathbb{Q}_p^3$, $u(\xi)$ is a unitary matrix,*

$$u(\xi)^{-1} = u(\xi)^* = \frac{(m+\lambda(\xi))I_4 - \beta\,\boldsymbol{\alpha}\cdot\mathbf{j}(\xi)}{\sqrt{2\lambda(\xi)(m+\lambda(\xi))}} \quad (\lambda(\xi)>0),$$

and

$$u(\xi)h(\xi)u(\xi)^{-1} = \beta\lambda(\xi). \tag{6.3}$$

Proof. If $\lambda(\xi)=0$, then $h(\xi)=0$ and $u(\xi)=I_4$, and there is nothing to prove. Assume $\lambda=\lambda(\xi)>0$, write $c=m+\lambda>0$, $N=\sqrt{2\lambda c}$, and $B=\beta\,\boldsymbol{\alpha}\cdot\mathbf{j}(\xi)$. By (4.1) and since $\mathbf{j}(\xi)$ is real, $B^*=(\boldsymbol{\alpha}\cdot\mathbf{j})\beta=-\beta\,\boldsymbol{\alpha}\cdot\mathbf{j}=-B$ and $B^2=-\beta^2(\boldsymbol{\alpha}\cdot\mathbf{j})^2=-|\mathbf{j}|^2 I_4$. Hence $u=(cI_4+B)/N$, $u^*=(cI_4-B)/N$, and

$$uu^* = u^*u = \frac{c^2I_4-B^2}{N^2} = \frac{(m+\lambda)^2+\lambda^2-m^2}{2\lambda(m+\lambda)}I_4 = I_4.$$

Next, using $\beta\,\boldsymbol{\alpha}\cdot\mathbf{j}\,\beta = -\boldsymbol{\alpha}\cdot\mathbf{j}$,

$$(cI_4+B)h = c\,\boldsymbol{\alpha}\cdot\mathbf{j} + cm\beta + \beta\,|\mathbf{j}|^2 - m\,\boldsymbol{\alpha}\cdot\mathbf{j} = \lambda\,\boldsymbol{\alpha}\cdot\mathbf{j} + \lambda(m+\lambda)\beta = \beta\lambda\,(cI_4+B),$$

where we used $c-m=\lambda$ and $cm+|\mathbf{j}|^2 = m^2+m\lambda+\lambda^2-m^2$. Thus $uh=\beta\lambda u$, which is (6.3). □

By using (6.1) and (6.3), the unitary transformation

$$\mathcal{W} := u\mathcal{F} : \mathfrak{H} \to \mathfrak{H}$$

(where u acts as the multiplication operator by the measurable, unitary-matrix-valued function $u(\xi)$) converts the p-adic Dirac operator $\boldsymbol{H}_0$ into a multiplication operator by the diagonal matrix $\beta\lambda(\xi)$,

$$\boldsymbol{H}_0 = \mathcal{W}^{-1}\beta\lambda(\xi)\mathcal{W} \quad \text{in } \mathfrak{H}. \tag{6.4}$$

Indeed, $\mathcal{W}\boldsymbol{H}_0\mathcal{W}^{-1} = u\mathcal{F}\boldsymbol{H}_0\mathcal{F}^{-1}u^{-1} = uhu^{-1} = \beta\lambda$.

6.3. **The spectrum of $\boldsymbol{H}_0$.** Since each $\mathfrak{j}_l(\xi_l)$ depends only on $|\xi_l|_p$, the function λ is constant on each of the sets

$$R_{\boldsymbol{\gamma}} := \left\{\xi \in \mathbb{Q}_p^3;\ |\xi_l|_p = p^{\gamma_l},\ l = 1,2,3\right\}, \qquad \boldsymbol{\gamma} = (\gamma_1,\gamma_2,\gamma_3) \in \mathbb{Z}^3,$$

which are open, compact, pairwise disjoint, have positive Haar measure, and cover $\mathbb{Q}_p^3$ up to the null set $\{\xi;\ \xi_1\xi_2\xi_3 = 0\}$. We denote by $\lambda_{\boldsymbol{\gamma}}$ the value of λ on $R_{\boldsymbol{\gamma}}$, and set $\Lambda := \{\lambda_{\boldsymbol{\gamma}};\ \boldsymbol{\gamma} \in \mathbb{Z}^3\}$, a countable subset of $[m, M]$, with M as in Proposition 6.1.

Theorem 6.1. *The free Dirac operator $\boldsymbol{H}_0$ is self-adjoint on $\mathfrak{H}$, and:*

(i) its spectrum $\sigma(\boldsymbol{H}_0)$ is the union of the essential ranges of the functions $\pm\lambda(\xi) : \mathbb{Q}_p^3 \to \mathbb{R}$, which equals the closure of $\Lambda \cup (-\Lambda)$ in $\mathbb{R}$; in particular, $\sigma(\boldsymbol{H}_0) \subset [-M,-m] \cup [m,M]$, and $\pm m \in \sigma(\boldsymbol{H}_0)$;

(ii) the spectrum of $\boldsymbol{H}_0$ is pure point: each $\pm\lambda_{\boldsymbol{\gamma}}$ is an eigenvalue of infinite multiplicity, and $\mathfrak{H}$ is the orthogonal direct sum of the corresponding eigenspaces; in particular, $\sigma(\boldsymbol{H}_0) = \sigma_{\text{ess}}(\boldsymbol{H}_0)$.

Proof. By (6.4), $\boldsymbol{H}_0$ is unitarily equivalent to the multiplication operator by $\beta\lambda(\xi)$ on $\mathfrak{H}$, i.e., to the direct sum of two copies of the multiplication by $\lambda(\xi)$ and two copies of the multiplication by $-\lambda(\xi)$ on $L^2(\mathbb{Q}_p^3)$. The spectrum of a multiplication operator is the essential range of the multiplier, see [46, Section VII.2], which gives the first assertion in (i). Since each $R_{\boldsymbol{\gamma}}$ has positive measure, every $\lambda_{\boldsymbol{\gamma}}$ belongs to the essential range of λ; conversely, λ takes values in Λ outside a null set, so its essential range is contained in $\overline{\Lambda}$; being closed, it equals $\overline{\Lambda}$. The bound $|\mathfrak{j}_l| \le 2\,\|J_l\|_1$ gives $m \le \lambda \le M$. Since $\mathfrak{j}_l$ is continuous with $\mathfrak{j}_l(0) = 0$, $\lambda(\xi) \to m$ as $\xi \to 0$, hence $m \in \overline{\Lambda}$. For (ii), the space $E^{\pm}_{\boldsymbol{\gamma}} := \mathcal{W}^{-1}\left(L^2(R_{\boldsymbol{\gamma}}) \otimes \frac{1}{2}(I_4 \pm \beta)\mathbb{C}^4\right)$ is infinite dimensional (because $R_{\boldsymbol{\gamma}}$ is open), and $\boldsymbol{H}_0\psi = \pm\lambda_{\boldsymbol{\gamma}}\psi$ for $\psi \in E^{\pm}_{\boldsymbol{\gamma}}$ by (6.4). Since the sets $R_{\boldsymbol{\gamma}}$ cover $\mathbb{Q}_p^3$ up to a null set, $\mathfrak{H} = \bigoplus_{\boldsymbol{\gamma},\pm} E^{\pm}_{\boldsymbol{\gamma}}$ (eigenspaces corresponding to the same value are merged). An eigenvalue of infinite multiplicity belongs to the essential spectrum, and the essential spectrum is closed, so $\sigma_{\text{ess}}(\boldsymbol{H}_0) = \sigma(\boldsymbol{H}_0)$. □

Remark 6.1. *We denote by $\sigma(\boldsymbol{H}_0^{Arch})$ the spectrum of the standard free Dirac operator, with the normalization $c = 1$; by [29, Theorem 1.1], $\sigma(\boldsymbol{H}_0^{Arch}) = (-\infty,-m] \cup [m,\infty)$. Then $\sigma(\boldsymbol{H}_0) \subset \sigma(\boldsymbol{H}_0^{Arch})$. However, in contrast with the standard case, where the spectrum is purely absolutely continuous, the spectrum of $\boldsymbol{H}_0$ is bounded and pure point. This is a*

consequence of the ultrametric structure of $\mathbb{Q}_p$: radial functions are locally constant away from the origin.

6.4. **Positive-, negative-, and zero-energy subspaces.** In the Hilbert space $\mathcal{W}\mathfrak{H}$, the p-adic Dirac operator is diagonal, see (6.4). Following [29, Section 1.4.2], and in order to include the massless case $m = 0$, we define the positive-, negative-, and zero-energy subspaces of $\boldsymbol{H}_0$ by means of the spectral theorem (functional calculus, [46, Section VII.2]):

$$\mathcal{P}_{pos} := 1_{(0,\infty)}(\boldsymbol{H}_0), \qquad \mathcal{P}_{neg} := 1_{(-\infty,0)}(\boldsymbol{H}_0), \qquad \mathcal{P}_0 := 1_{\{0\}}(\boldsymbol{H}_0),$$

and $\mathfrak{H}_{pos} := \mathcal{P}_{pos}\mathfrak{H}$, $\mathfrak{H}_{neg} := \mathcal{P}_{neg}\mathfrak{H}$, $\mathfrak{H}_0^{\ker} := \mathcal{P}_0\mathfrak{H} = \ker \boldsymbol{H}_0$. These are mutually orthogonal projections with $\mathcal{P}_{pos} + \mathcal{P}_{neg} + \mathcal{P}_0 = 1$ and they commute with $\boldsymbol{H}_0$, so

$$\mathfrak{H} = \mathfrak{H}_{pos} \oplus \mathfrak{H}_0^{\ker} \oplus \mathfrak{H}_{neg}, \tag{6.5}$$

and each summand is invariant under $\boldsymbol{H}_0$ and under $e^{-\mathrm{i}t\boldsymbol{H}_0}$. By (6.4), with $Z_0 := \{\xi \in \mathbb{Q}_p^3;\ \lambda(\xi) = 0\}$,

$$\mathcal{P}_{\substack{pos\\neg}} = \mathcal{W}^{-1}\, \tfrac{1}{2}(I_4 \pm \beta)\, 1_{\mathbb{Q}_p^3 \smallsetminus Z_0}(\xi)\, \mathcal{W}, \qquad \mathcal{P}_0 = \mathcal{F}^{-1} 1_{Z_0}(\xi) \mathcal{F}, \tag{6.6}$$

where in the last formula we used $u(\xi) = I_4$ on Z_0. For $\psi \in \mathfrak{H}_{pos}$, $\phi_+ := \mathcal{W}\psi$ satisfies $(\psi, \boldsymbol{H}_0\psi) = (\phi_+, \lambda(\xi)\phi_+) \geq 0$, with equality only if $\psi = 0$; similarly $(\psi, \boldsymbol{H}_0\psi) < 0$ for $0 \neq \psi \in \mathfrak{H}_{neg}$.

Let $|\boldsymbol{H}_0| := \sqrt{\boldsymbol{H}_0^2}$, which is the pseudo-differential operator with symbol $\lambda(\xi) I_4$, i.e., $|\boldsymbol{H}_0| = \sqrt{\boldsymbol{J}_{x_1}^2 + \boldsymbol{J}_{x_2}^2 + \boldsymbol{J}_{x_3}^2 + m^2}$, and $\operatorname{sgn} \boldsymbol{H}_0 := \mathcal{P}_{pos} - \mathcal{P}_{neg}$, which is the pseudo-differential operator with symbol $\lambda(\xi)^{-1} h(\xi)$ on $\mathbb{Q}_p^3 \smallsetminus Z_0$ and 0 on Z_0. Then $\boldsymbol{H}_0 = |\boldsymbol{H}_0| \operatorname{sgn} \boldsymbol{H}_0$, which is the polar decomposition of $\boldsymbol{H}_0$, and $\boldsymbol{H}_0\psi = \pm\, |\boldsymbol{H}_0|\, \psi$ for $\psi \in \mathfrak{H}_{\substack{pos\\neg}}$.

The massive case. If $m > 0$, then $\lambda \geq m > 0$, $Z_0 = \emptyset$, $\mathcal{P}_0 = 0$, $|\boldsymbol{H}_0| \geq m$ is invertible, $\operatorname{sgn} \boldsymbol{H}_0 = \boldsymbol{H}_0\, |\boldsymbol{H}_0|^{-1}$, and we recover the familiar formulas $\mathcal{P}_{\substack{pos\\neg}} = \frac{1}{2}\left(1 \pm \boldsymbol{H}_0\, |\boldsymbol{H}_0|^{-1}\right)$ and $\mathfrak{H} = \mathfrak{H}_{pos} \oplus \mathfrak{H}_{neg}$.

The massless case. If $m = 0$, then $\lambda(\xi) = |\mathfrak{j}(\xi)|$, $0 \in \sigma(\boldsymbol{H}_0)$ by Theorem 6.1, and the operator $|\boldsymbol{H}_0|$ is not invertible (its inverse would be unbounded, because $\lambda(\xi) \to 0$ as $\xi \to 0$). The expression $\boldsymbol{H}_0 / |\boldsymbol{H}_0|$ is therefore not defined as a bounded operator, and the decomposition (6.5) must include the kernel of $\boldsymbol{H}_0$.

Proposition 6.2. *Assume $m = 0$, and set $Z^{(l)} := \{\xi_l \in \mathbb{Q}_p;\ \mathfrak{j}_l(\xi_l) = 0\}$, so that $Z_0 = Z^{(1)} \times Z^{(2)} \times Z^{(3)}$. Then*

$$\ker \boldsymbol{H}_0 = \left\{\psi \in \mathfrak{H};\ \widehat{\psi}(\xi) = 0 \text{ for almost every } \xi \notin Z_0\right\}.$$

Moreover:

- *(i) if every $Z^{(l)}$ has positive Haar measure, then $\ker \boldsymbol{H}_0$ is infinite dimensional, and 0 is an eigenvalue of infinite multiplicity of $\boldsymbol{H}_0$. This is the case if each J_l is supported in a ball $p^{-L_l}\mathbb{Z}_p$: then $Z_0 \supset p^{L_1}\mathbb{Z}_p \times p^{L_2}\mathbb{Z}_p \times p^{L_3}\mathbb{Z}_p$, and $\ker \boldsymbol{H}_0$ contains every $\psi \in \mathfrak{H}$ that is invariant under the translations by $p^{-L_1}\mathbb{Z}_p \times p^{-L_2}\mathbb{Z}_p \times p^{-L_3}\mathbb{Z}_p$;*

(ii) if some J_l satisfies $J_l \geq 0$ and $J_l(p^k) > 0$ for infinitely many $k > 0$, then $Z^{(l)} = \{0\}$, $\ker \boldsymbol{H}_0 = \{0\}$, but $0 \in \sigma(\boldsymbol{H}_0)$ is an accumulation point of eigenvalues.

Proof. Since $h(\xi)^2 = \lambda(\xi)^2 I_4$, $h(\xi)$ is invertible for $\xi \notin Z_0$ and vanishes for $\xi \in Z_0$; hence $\boldsymbol{H}_0\psi = 0$ if and only if $h(\xi)\widehat{\psi}(\xi) = 0$ a.e., if and only if $\widehat{\psi} = 0$ a.e. outside Z_0. (i) If Z_0 has positive measure, it contains a set $R_{\boldsymbol{\gamma}}$ (because λ is constant on these sets), and $\mathcal{F}^{-1}(L^2(R_{\boldsymbol{\gamma}}) \otimes \mathbb{C}^4)$ is infinite dimensional. If J_l is supported in $p^{-L_l}\mathbb{Z}_p$, then $\mathfrak{j}_l(\xi_l) = 0$ for $|\xi_l|_p \leq p^{-L_l}$ by Example 4.1. A function $\psi \in L^2$ is invariant under translations by a compact open subgroup G if and only if $\widehat{\psi}$ is supported in the annihilator $\{\xi;\ \chi_p(\xi \cdot a) = 1\ \forall a \in G\}$, which for $G = \prod_l p^{-L_l}\mathbb{Z}_p$ is $\prod_l p^{L_l}\mathbb{Z}_p$. (ii) For $\xi_l \neq 0$, $\widehat{J_l}(0) - \widehat{J_l}(|\xi_l|_p) = \int_{\mathbb{Q}_p} J_l(|y|_p)\{1 - \operatorname{Re}\chi_p(\xi_l y)\}dy$. The integrand is non-negative. Choose $k > 0$ with $J_l(p^k) > 0$ and $p^k |\xi_l|_p \geq p$; on the sphere $|y|_p = p^k$ the function $y \mapsto \chi_p(\xi_l y)$ is not identically 1 (its integral over the sphere is 0 or $-p^{k-1}$, whereas the sphere has measure $p^k(1 - p^{-1})$), so the integral is strictly positive and $\mathfrak{j}_l(\xi_l) \neq 0$. Thus $Z^{(l)} = \{0\}$ has measure zero, and $\ker \boldsymbol{H}_0 = \{0\}$. The last assertion follows from Theorem 6.1(i). □

Remark 6.2. *In the setting of Section 9 (kernels supported in $\mathbb{Z}_p$), the zero-energy subspace of the massless operator contains the functions on $\mathbb{Z}_p^3$ that are constant, tensored with $\mathbb{C}^4$; on the finite graph G_n^3 this corresponds to the uniform distribution over the vertices.*

Again, following the classical case, we define the Foldy–Wouthuysen transformation as $\mathcal{U}_{FW} = \mathcal{F}^{-1}\mathcal{W} = \mathcal{F}^{-1}u\mathcal{F}$. It transforms the free Dirac operator into the pseudo-differential operator

$$\begin{aligned}\mathcal{U}_{FW}\boldsymbol{H}_0\mathcal{U}_{FW}^{-1} &= \mathcal{F}^{-1}\, uhu^{-1}\, \mathcal{F} \\ &= \begin{bmatrix} \sqrt{\boldsymbol{J}_{x_1}^2 + \boldsymbol{J}_{x_2}^2 + \boldsymbol{J}_{x_3}^2 + m^2} & \mathbf{0} \\ \mathbf{0} & -\sqrt{\boldsymbol{J}_{x_1}^2 + \boldsymbol{J}_{x_2}^2 + \boldsymbol{J}_{x_3}^2 + m^2} \end{bmatrix} = \beta\,|\boldsymbol{H}_0|\,.\end{aligned}$$

We interpret this formula as the fact that the free Dirac equation is unitarily equivalent to a pair of (two-component) square-root Klein–Gordon equations.

7. Charge conjugation

Following the standard case, see [29, Section 1.4.6], the p-adic Dirac operator for a charge $e \in \mathbb{R}$ in an external electromagnetic field is given by

$$\boldsymbol{H}(e) = \boldsymbol{H}(e; \boldsymbol{A}, \varphi) := \boldsymbol{\alpha} \cdot (\nabla - e\boldsymbol{A}(t, x)) + \beta m + e\varphi(t, x) I_4,$$

where the scalar potential $\varphi : \mathbb{R} \times \mathbb{Q}_p^3 \to \mathbb{R}$ and the vector potential $\boldsymbol{A} = (A_1, A_2, A_3) : \mathbb{R} \times \mathbb{Q}_p^3 \to \mathbb{R}^3$ are bounded measurable functions, acting as multiplication operators. Thus, for every t, $\boldsymbol{H}(e)$ is a bounded self-adjoint operator on $\mathfrak{H}$.

In the standard case, the charge conjugation is the antiunitary map $\Psi \mapsto U\overline{\Psi}$ with $U = \mathrm{i}\beta\alpha_2$ (up to a phase). That choice relies on the identity $\overline{-\mathrm{i}\partial_{x_l}\Psi} = \mathrm{i}\partial_{x_l}\overline{\Psi}$. In our setting,

the kernels J_l are real, so $\overline{\boldsymbol{J}_{x_l}\Psi} = \boldsymbol{J}_{x_l}\overline{\Psi}$ (Remark 4.2), and the matrix U has to be chosen differently. We define the charge conjugation $\mathcal{C}$ as the antiunitary transformation

$$\mathcal{C}\Psi := U_{\mathcal{C}}\overline{\Psi}, \qquad U_{\mathcal{C}} := \alpha_2.$$

Since α_2 is unitary and $\overline{\alpha_2} = -\alpha_2$, we have $\mathcal{C}^2\Psi = \alpha_2\overline{\alpha_2}\Psi = -\alpha_2^2\Psi = -\Psi$, that is, $\mathcal{C}^2 = -1$.

Lemma 7.1. *With the above notation, the following assertions hold.*

(i) $\mathcal{C}\boldsymbol{J}_{x_l} = \boldsymbol{J}_{x_l}\mathcal{C}$, $\mathcal{C}\alpha_l = -\alpha_l\mathcal{C}$ *for* $l = 1, 2, 3$, *and* $\mathcal{C}\beta = -\beta\mathcal{C}$, *as operators on* $\mathfrak{H}$. *Moreover, up to a phase factor,* $U_{\mathcal{C}} = \alpha_2$ *is the only* 4×4 *matrix* U *such that* $U\overline{\alpha_l}U^{-1} = -\alpha_l$, $l = 1, 2, 3$, *and* $U\beta U^{-1} = -\beta$.

(ii) $\mathcal{C}\boldsymbol{H}_0\mathcal{C}^{-1} = -\boldsymbol{H}_0$.

(iii) $\mathcal{C}\boldsymbol{H}(e; \boldsymbol{A}, \varphi)\mathcal{C}^{-1} = -\boldsymbol{H}(-e; -\boldsymbol{A}, \varphi)$. *In particular, in a purely electrostatic field (*$\boldsymbol{A} = 0$*),* $\mathcal{C}\boldsymbol{H}(e)\mathcal{C}^{-1} = -\boldsymbol{H}(-e)$.

(iv) If $\Psi \in C^1(\mathbb{R}, \mathfrak{H})$ *is a solution of*

$$\mathrm{i}\frac{\partial}{\partial t}\Psi(t, x) = \boldsymbol{H}(e; \boldsymbol{A}, \varphi)\Psi(t, x), \tag{7.1}$$

then

$$\mathrm{i}\frac{\partial}{\partial t}\mathcal{C}\Psi(t, x) = \boldsymbol{H}(-e; -\boldsymbol{A}, \varphi)\,\mathcal{C}\Psi(t, x).$$

Proof. (i) Since J_l is real valued, $\overline{\boldsymbol{J}_{x_l}\Psi} = \boldsymbol{J}_{x_l}\overline{\Psi}$, and $\boldsymbol{J}_{x_l}$ acts componentwise, so it commutes with the matrix α_2; hence $\mathcal{C}\boldsymbol{J}_{x_l} = \boldsymbol{J}_{x_l}\mathcal{C}$. Next, $\mathcal{C}\alpha_l\Psi = \alpha_2\overline{\alpha_l}\,\overline{\Psi}$. By (4.1), $\alpha_2\overline{\alpha_l} = \alpha_2\alpha_l = -\alpha_l\alpha_2$ for $l = 1, 3$, and $\alpha_2\overline{\alpha_2} = -\alpha_2\alpha_2$; hence $\mathcal{C}\alpha_l = -\alpha_l\mathcal{C}$ for $l = 1, 2, 3$. Similarly $\alpha_2\overline{\beta} = \alpha_2\beta = -\beta\alpha_2$. For the uniqueness, the conditions on U are equivalent to $U\alpha_1 = -\alpha_1 U$, $U\alpha_3 = -\alpha_3 U$, $U\beta = -\beta U$, $U\alpha_2 = \alpha_2 U$. The 16 products $\Gamma_S = \prod_{g\in S} g$, $S \subset \{\alpha_1, \alpha_2, \alpha_3, \beta\}$ (in a fixed order), form a basis of the space of 4×4 complex matrices, and, by (4.1), each generator g satisfies $g\Gamma_S = \pm\Gamma_S g$, with sign $(-1)^{|S|-1}$ if $g \in S$ and $(-1)^{|S|}$ if $g \notin S$. Writing $U = \sum_S c_S\Gamma_S$, the four conditions force $c_S = 0$ unless Γ_S anticommutes with $\alpha_1, \alpha_3, \beta$ and commutes with α_2; if $|S|$ is odd, this means $S = \{\alpha_2\}$, and if $|S|$ is even, this means $S = \{\alpha_1, \alpha_3, \beta\}$, which has odd cardinality, a contradiction. Hence $U = c\,\alpha_2$.

(ii) By (i), $\mathcal{C}\boldsymbol{H}_0 = \sum_l \mathcal{C}\alpha_l\boldsymbol{J}_{x_l} + m\mathcal{C}\beta = -\sum_l \alpha_l\boldsymbol{J}_{x_l}\mathcal{C} - m\beta\mathcal{C} = -\boldsymbol{H}_0\mathcal{C}$.

(iii) The functions A_l, φ are real valued, so the corresponding multiplication operators commute with $\mathcal{C}$. By (i),

$$\begin{aligned}\mathcal{C}\boldsymbol{H}(e; \boldsymbol{A}, \varphi)\mathcal{C}^{-1} &= -\boldsymbol{\alpha}\cdot\nabla + e\boldsymbol{\alpha}\cdot\boldsymbol{A} - \beta m + e\varphi I_4 \\ &= -\{\boldsymbol{\alpha}\cdot(\nabla - (-e)(-\boldsymbol{A})) + \beta m + (-e)\varphi I_4\} = -\boldsymbol{H}(-e; -\boldsymbol{A}, \varphi).\end{aligned}$$

(iv) Apply $\mathcal{C}$ to (7.1). Since $\mathcal{C}$ is antilinear and continuous, $\mathcal{C}(\mathrm{i}\partial_t\Psi) = -\mathrm{i}\partial_t\mathcal{C}\Psi$, and by (iii), $-\mathrm{i}\partial_t\mathcal{C}\Psi = \mathcal{C}\boldsymbol{H}(e; \boldsymbol{A}, \varphi)\Psi = -\boldsymbol{H}(-e; -\boldsymbol{A}, \varphi)\mathcal{C}\Psi$. □

Remark 7.1. *(1) In the standard case,* $\mathcal{C}\boldsymbol{H}(e)\mathcal{C}^{-1} = -\boldsymbol{H}(-e)$ *holds with the* same *vector potential, and* $\mathcal{C}^2 = 1$. *Both differences come from the fact that the momentum-type operators* $\boldsymbol{J}_{x_l}$ *are invariant under complex conjugation, as are the multiplication operators* A_l*: in the p-adic model, the kinetic term* $\boldsymbol{\alpha}\cdot\nabla$ *and the magnetic coupling* $\boldsymbol{\alpha}\cdot\boldsymbol{A}$ *transform in the same*

way, so a symmetry that reverses the sign of the former must also reverse the sign of the latter. In the presence of a vector potential, the symmetry $\mathcal{C}$ thus combines charge conjugation with the reversal $\boldsymbol{A} \mapsto -\boldsymbol{A}$ (as happens, in the standard theory, under time reversal). By the uniqueness part of Lemma 7.1(i), this cannot be remedied by choosing a different matrix $U_{\mathcal{C}}$ while keeping the operators $\boldsymbol{J}_{x_l}$.

(2) In [10], charge conjugation was established for a p-adic Dirac equation built from the Vladimirov operator; that result cannot be transferred to the present setting, since the operators $\boldsymbol{J}_{x_l}$ are different. In particular, the matrix $-\mathrm{i}\beta\alpha_2$ does not implement a charge-conjugation symmetry of $\boldsymbol{H}_0$: one checks that $(-\mathrm{i}\beta\alpha_2)\overline{\boldsymbol{H}_0}(-\mathrm{i}\beta\alpha_2)^{-1} = \boldsymbol{\alpha}\cdot\nabla - \beta m \neq -\boldsymbol{H}_0$.

Proposition 7.1. *The charge conjugation $\mathcal{C}$ satisfies $\mathcal{C}\mathcal{P}_{pos}\mathcal{C}^{-1} = \mathcal{P}_{neg}$, $\mathcal{C}\mathcal{P}_{neg}\mathcal{C}^{-1} = \mathcal{P}_{pos}$, and $\mathcal{C}\mathcal{P}_0\mathcal{C}^{-1} = \mathcal{P}_0$. Consequently, $\mathcal{C}$ maps $\mathfrak{H}_{neg}$ onto $\mathfrak{H}_{pos}$, $\mathfrak{H}_{pos}$ onto $\mathfrak{H}_{neg}$, and $\ker \boldsymbol{H}_0$ onto itself. Moreover, $|\mathcal{C}\psi(x)|^2 = |\psi(x)|^2$ for every $\psi \in \mathfrak{H}$ and almost every $x \in \mathbb{Q}_p^3$.*

Proof. By Lemma 7.1(ii), $\mathcal{C}\boldsymbol{H}_0^k\mathcal{C}^{-1} = (-\boldsymbol{H}_0)^k$ for every k, so, for every polynomial P with real coefficients, $\mathcal{C}P(\boldsymbol{H}_0)\mathcal{C}^{-1} = P(-\boldsymbol{H}_0)$. By the Weierstrass theorem on the compact set $\sigma(\boldsymbol{H}_0) \cup (-\sigma(\boldsymbol{H}_0))$ and the continuity of the functional calculus, $\mathcal{C}f(\boldsymbol{H}_0)\mathcal{C}^{-1} = f(-\boldsymbol{H}_0)$ for every real continuous function f, and then, by monotone (strong) limits, for every real bounded Borel function f. Taking $f = 1_{(0,\infty)}$, $1_{(-\infty,0)}$, $1_{\{0\}}$ gives the first assertions. The last one follows from the fact that α_2 is unitary. □

Hence, in the free case and in the electrostatic case, the negative-energy subspace of $\boldsymbol{H}(e)$ is connected via a symmetry transformation with the positive-energy subspace of the Dirac operator $\boldsymbol{H}(-e)$ for a particle with opposite charge (antiparticle). For $\mathcal{C}\psi$ in the positive-energy subspace of $\boldsymbol{H}(-e)$, by interpreting $|\mathcal{C}\psi(x)|^2$ as a position probability density, the equality $|\mathcal{C}\psi(x)|^2 = |\psi(x)|^2$ shows that the motion of a negative-energy state ψ is indistinguishable from that of a positive-energy antiparticle. Then, as in the standard case, one obtains the interpretation:

a state $\psi \in \mathfrak{H}_{neg}$ describes an antiparticle with positive energy.

In the presence of a vector potential, by Lemma 7.1(iii), the same interpretation holds provided the antiparticle is subject to the reversed vector potential $-\boldsymbol{A}$.

8. Continuous-time quantum walks I

In this section, we discretize the p-adic Dirac evolution on a countable graph. We first describe what can be said for an arbitrary bounded self-adjoint Hamiltonian (Section 8.2), and then show that, for $\boldsymbol{H}_0$ and arbitrary integrable radial kernels J_1, J_2, J_3, one obtains a CTQW in the strict sense (Section 8.3). To avoid a conflict with the index l of the spatial directions, we use n to denote the discretization level.

8.1. The countable graph $\widetilde{G}_n^3$ and the basis $\psi_{\boldsymbol{j},b}$.

For an integer $n \geq 1$, set

$$\widetilde{G}_n := \mathbb{Q}_p/p^n\mathbb{Z}_p, \qquad \widetilde{G}_n^3 := \mathbb{Q}_p^3/p^n\mathbb{Z}_p^3 = \left(\mathbb{Q}_p/p^n\mathbb{Z}_p\right)^3.$$

We use a set of representatives of $\widetilde{G}_n$ of the form $i=i_{-r}p^{-r}+i_{-r+1}p^{-r+1}+\ldots+i_0+\ldots+i_{n-1}p^{n-1}$, with $r\geq 0$ and digits $i_k\in\{0,1,\ldots,p-1\}$. The elements of $\widetilde{G}_n^3$ are denoted as $\boldsymbol{i}=(i_1,i_2,i_3)$. Notice that $\widetilde{G}_n^3$ is a countable set, and that, for $i\neq k$ in $\widetilde{G}_n$, the number $|i-k|_p>p^{-n}$ does not depend on the choice of representatives. We consider the partition

$$\mathbb{Q}_p^3=\bigsqcup_{\boldsymbol{i}\in\widetilde{G}_n^3}\left(\boldsymbol{i}+p^n\mathbb{Z}_p^3\right), \tag{8.1}$$

where the balls $\boldsymbol{i}+p^n\mathbb{Z}_p^3$ are pairwise disjoint. We denote by $\Omega(p^n\left\|x-\boldsymbol{i}\right\|_p)$ the characteristic function of the ball $\boldsymbol{i}+p^n\mathbb{Z}_p^3$, by $e_1,\ldots,e_4$ the standard basis of $\mathbb{C}^4$, and set

$$\psi_{\boldsymbol{j},b}(x):=p^{\frac{3n}{2}}\Omega(p^n\left\|x-\boldsymbol{j}\right\|_p)\,e_b\in\mathfrak{H},\qquad \boldsymbol{j}\in\widetilde{G}_n^3,\ b=1,2,3,4. \tag{8.2}$$

Since $\int\Omega(p^n\left\|x-\boldsymbol{j}\right\|_p)d^3x=p^{-3n}$ and the balls are disjoint, $\{\psi_{\boldsymbol{j},b}\}$ is an orthonormal set in $\mathfrak{H}$. Its closed linear span is $V_n\otimes\mathbb{C}^4$, where

$$V_n:=\left\{\phi\in L^2(\mathbb{Q}_p^3);\ \phi(x+a)=\phi(x)\text{ for all }a\in p^n\mathbb{Z}_p^3\right\}$$

is the space of square-integrable functions which are constant on the balls of radius p^{-n}.

8.2. **Coarse-grained position probabilities for a general Hamiltonian.** Let $\boldsymbol{H}$ be a bounded self-adjoint operator on $\mathfrak{H}$, not necessarily $\boldsymbol{H}_0$; so $\{e^{-\mathrm{i}t\boldsymbol{H}}\}_{t\in\mathbb{R}}$ is a group of unitary operators on $\mathfrak{H}$. For $\boldsymbol{i}\in\widetilde{G}_n^3$, let $P_{\boldsymbol{i}}$ be the orthogonal projection on $\mathfrak{H}$ given by multiplication by $\Omega(p^n\left\|x-\boldsymbol{i}\right\|_p)$; by (8.1), $\{P_{\boldsymbol{i}}\}_{\boldsymbol{i}}$ is a projection-valued measure (a resolution of the identity: $P_{\boldsymbol{i}}P_{\boldsymbol{k}}=\delta_{\boldsymbol{i},\boldsymbol{k}}P_{\boldsymbol{i}}$, $\sum_{\boldsymbol{i}}P_{\boldsymbol{i}}=1$ strongly). It describes the measurement of the position of the particle with resolution p^{-n}. Set

$$\Psi_{\boldsymbol{j},b}(t,x):=e^{-\mathrm{i}t\boldsymbol{H}}\psi_{\boldsymbol{j},b}(x)=\left[\Psi_{\boldsymbol{j},b}^{(1)}(t,x),\ldots,\Psi_{\boldsymbol{j},b}^{(4)}(t,x)\right]^T,$$

so that $\|\Psi_{\boldsymbol{j},b}(t,\cdot)\|=1$ for all t, and

$$\widetilde{\pi}_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t):=\|P_{\boldsymbol{i}}\Psi_{\boldsymbol{j},b}(t,\cdot)\|^2=\sum_{a=1}^{4}\int_{\boldsymbol{i}+p^n\mathbb{Z}_p^3}\left|\Psi_{\boldsymbol{j},b}^{(a)}(t,x)\right|^2d^3x.$$

By the Born rule, $\widetilde{\pi}_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t)$ is the probability that a measurement of the position, with resolution p^{-n}, performed at time t, finds the particle in the ball $\boldsymbol{i}+p^n\mathbb{Z}_p^3$, given that at time 0 the particle was prepared in the state $\psi_{\boldsymbol{j},b}$, uniformly localized in the ball $\boldsymbol{j}+p^n\mathbb{Z}_p^3$ with internal state e_b. Using (8.1),

$$\sum_{\boldsymbol{i}\in\widetilde{G}_n^3}\widetilde{\pi}_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t)=\sum_{a=1}^{4}\int_{\mathbb{Q}_p^3}\left|\Psi_{\boldsymbol{j},b}^{(a)}(t,x)\right|^2d^3x=\|\Psi_{\boldsymbol{j},b}(t,\cdot)\|^2=1,$$

so, for every t and b, $[\widetilde{\pi}_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t)]_{\boldsymbol{i},\boldsymbol{j}}$ is a (column) stochastic matrix indexed by the countable set $\widetilde{G}_n^3$.

These numbers should be distinguished from the transition probabilities

$$\pi_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t):=\left|(e^{-\mathrm{i}t\boldsymbol{H}}\psi_{\boldsymbol{j},b},\psi_{\boldsymbol{i},a})\right|^2$$

from the state $\psi_{\boldsymbol{j},b}$ to the state $\psi_{\boldsymbol{i},a}$, which are the ones used in the theory of CTQWs [39]. Since $\{p^{3n/2}\Omega(p^n \|x-\boldsymbol{i}\|_p)e_a\}_a$ is an orthonormal set in the range of $P_{\boldsymbol{i}}$, Bessel's inequality gives

$$\widetilde{\pi}^{(b)}_{\boldsymbol{i},\boldsymbol{j}}(t) \geq \sum_{a=1}^{4} \pi_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t), \tag{8.3}$$

with equality for all $\boldsymbol{i}$ if and only if $\Psi_{\boldsymbol{j},b}(t,\cdot) \in V_n \otimes \mathbb{C}^4$. The difference is the probability carried by the components of the wavefunction which vary inside the balls of radius p^{-n}. Consequently, for a general $\boldsymbol{H}$, the coarse-grained dynamics $t \mapsto \widetilde{\pi}^{(b)}(t)$ is *not* a CTQW on $\widetilde{G}^3_n$ in the strict sense: there is, in general, no Hamiltonian on $\ell^2(\widetilde{G}^3_n)\otimes\mathbb{C}^4$ generating it. It is a CTQW exactly when $V_n \otimes \mathbb{C}^4$ is invariant under $e^{-\mathrm{i}t\boldsymbol{H}}$. We now show that this is the case for $\boldsymbol{H}_0$.

8.3. **An exact CTQW for the free p-adic Dirac equation.**

Lemma 8.1. *Let J_1, J_2, J_3 be arbitrary integrable radial kernels as in Section 4.3. Then V_n is invariant under $\boldsymbol{J}_{x_1}$, $\boldsymbol{J}_{x_2}$, $\boldsymbol{J}_{x_3}$, and $V_n \otimes \mathbb{C}^4$ is invariant under $\boldsymbol{H}_0$ and under $e^{-\mathrm{i}t\boldsymbol{H}_0}$, $t \in \mathbb{R}$. Moreover, for $l = 1,2,3$ and $\phi \in V_n$, writing $\phi(\boldsymbol{i})$ for the value of ϕ on the ball $\boldsymbol{i} + p^n\mathbb{Z}^3_p$,*

$$(\boldsymbol{J}_{x_l}\phi)(\boldsymbol{i}) = \sum_{k\in\widetilde{G}_n,\, k\neq i_l} w_l(i_l,k)\left\{\phi(\boldsymbol{i}^{(l)}(k)) - \phi(\boldsymbol{i})\right\}, \qquad w_l(i,k) := p^{-n}J_l(|i-k|_p), \tag{8.4}$$

where $\boldsymbol{i}^{(l)}(k)$ is obtained from $\boldsymbol{i}$ by replacing its l-th coordinate by k. The series converges absolutely, since $\sum_{k\neq i}|w_l(i,k)| = \int_{|y|_p>p^{-n}}\left|J_l(|y|_p)\right|dy \leq \|J_l\|_1$.

Proof. A function $\phi \in L^2(\mathbb{Q}^3_p)$ belongs to V_n if and only if $(\chi_p(a\cdot\xi)-1)\widehat{\phi}(\xi) = 0$ for all $a \in p^n\mathbb{Z}^3_p$, i.e., if and only if $\widehat{\phi}$ is supported in the annihilator $p^{-n}\mathbb{Z}^3_p$ of $p^n\mathbb{Z}^3_p$. Since $\boldsymbol{J}_{x_l}$ is the Fourier multiplier with symbol $\mathfrak{j}_l(\xi_l)$, it preserves this support condition; hence $\boldsymbol{J}_{x_l}V_n \subset V_n$, and therefore $\boldsymbol{H}_0(V_n\otimes\mathbb{C}^4) \subset V_n\otimes\mathbb{C}^4$. Since $\boldsymbol{H}_0$ is bounded, $e^{-\mathrm{i}t\boldsymbol{H}_0} = \sum_{\nu\geq 0}(-\mathrm{i}t\boldsymbol{H}_0)^\nu/\nu!$ also preserves the closed subspace $V_n\otimes\mathbb{C}^4$. To compute (8.4), it is enough to consider one variable. If $x \in i + p^n\mathbb{Z}_p$ and $y \in k + p^n\mathbb{Z}_p$ with $k \neq i$, then $|x-y|_p = |i-k|_p$ by the ultrametric property. Hence, for $\phi \in V_n$ (one variable),

$$\begin{aligned}\boldsymbol{J}\phi(x) &= \sum_{k\neq i}\int_{k+p^n\mathbb{Z}_p} J(|x-y|_p)\{\phi(k)-\phi(i)\}dy + \int_{i+p^n\mathbb{Z}_p} J(|x-y|_p)\{\phi(i)-\phi(i)\}dy \\ &= \sum_{k\neq i} p^{-n}J(|i-k|_p)\{\phi(k)-\phi(i)\}.\end{aligned}$$

The bound on $\sum_k |w(i,k)|$ follows from $\{y;\ |y|_p > p^{-n}\} = \bigsqcup_{k\neq 0}(k+p^n\mathbb{Z}_p)$. □

Formula (8.4) says that, on $V_n \cong \ell^2(\widetilde{G}^3_n)$, the operator $\boldsymbol{J}_{x_l}$ is the weighted graph Laplacian $\boldsymbol{L}^{(l)} := A^{(l)} - D^{(l)}$ (with the sign convention of (8.4)) of the graph with vertex set $\widetilde{G}^3_n$, in which $\boldsymbol{i}$ and $\boldsymbol{k}$ are joined by an edge if they differ only in the l-th coordinate, with weight $w_l(i_l,k_l)$; here $A^{(l)}$ is the weighted adjacency matrix and $D^{(l)} = d_l\mathrm{Id}$, with $d_l := \int_{|y|_p>p^{-n}} J_l(|y|_p)dy$,

is the (constant) weighted degree. The weights depend only on $|i_l-k_l|_p$, that is, on the level of the p-adic tree at which the branches containing i_l and k_l separate: this is the hierarchical (ultrametric) structure of the graph. The matrix of $\boldsymbol{L}^{(l)}$ in the orthonormal basis $\{p^{3n/2}\Omega(p^n\|x-\boldsymbol{i}\|_p)\}_{\boldsymbol{i}}$ is real and symmetric, and $\left\|\boldsymbol{L}^{(l)}\right\|\leq 2\|J_l\|_1$.

Theorem 8.1. *Let J_1,J_2,J_3 be arbitrary integrable radial kernels, and let $\mathcal{I}_n:\ell^2(\widetilde{G}_n^3)\otimes\mathbb{C}^4\to V_n\otimes\mathbb{C}^4\subset\mathfrak{H}$ be the unitary map sending the canonical basis vector $\delta_{\boldsymbol{j}}\otimes e_b$ to $\psi_{\boldsymbol{j},b}$. Then*

$$\mathcal{I}_n^{-1}\,\boldsymbol{H}_0\,\mathcal{I}_n=\boldsymbol{H}_n:=\sum_{l=1}^{3}\boldsymbol{L}^{(l)}\otimes\alpha_l+m\,\mathrm{Id}\otimes\beta, \tag{8.5}$$

a bounded self-adjoint operator on $\ell^2(\widetilde{G}_n^3)\otimes\mathbb{C}^4$, and $\mathcal{I}_n^{-1}e^{-\mathrm{i}t\boldsymbol{H}_0}\mathcal{I}_n=e^{-\mathrm{i}t\boldsymbol{H}_n}$ for every $t\in\mathbb{R}$. Consequently, the restriction of the free p-adic Dirac evolution to $V_n\otimes\mathbb{C}^4$ is the continuous-time quantum walk on the weighted graph $\widetilde{G}_n^3$, with internal space $\mathbb{C}^4$ at each vertex, generated by $\boldsymbol{H}_n$; and, for $\boldsymbol{H}=\boldsymbol{H}_0$, equality holds in (8.3):

$$\widetilde{\pi}_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t)=\sum_{a=1}^{4}\left|\left(e^{-\mathrm{i}t\boldsymbol{H}_n}\right)_{(\boldsymbol{i},a),(\boldsymbol{j},b)}\right|^2.$$

Proof. This follows from Lemma 8.1, since $\boldsymbol{H}_0$ acts on $V_n\otimes\mathbb{C}^4$ as $\sum_l\boldsymbol{J}_{x_l}\alpha_l+m\beta$, and $\mathcal{I}_n$ intertwines $\boldsymbol{J}_{x_l}|_{V_n}$ with $\boldsymbol{L}^{(l)}$. The equality in (8.3) holds because $e^{-\mathrm{i}t\boldsymbol{H}_0}\psi_{\boldsymbol{j},b}\in V_n\otimes\mathbb{C}^4$. □

Remark 8.1. *The CTQW of Theorem 8.1 is an exact discretization: no approximation is involved, because $V_n\otimes\mathbb{C}^4$ is a reducing subspace of $\boldsymbol{H}_0$ (indeed, the orthogonal projection onto it is the Fourier multiplier $1_{p^{-n}\mathbb{Z}_p^3}(\xi)$, which commutes with $\boldsymbol{H}_0$). The spaces V_n increase with n, and $\bigcup_n V_n$ is dense in $L^2(\mathbb{Q}_p^3)$, so the walks for all n together determine the full evolution $e^{-\mathrm{i}t\boldsymbol{H}_0}$.*

9. Continuous-time quantum walks II

9.1. **Preliminary results.** In this section, we work with the p-adic Dirac Hamiltonian $\boldsymbol{H}_0$, but select the kernels J_1, J_2, J_3 such that the condition

$$e^{-\mathrm{i}t\boldsymbol{H}_0}\left(L^2(\mathbb{Z}_p^3)\otimes\mathbb{C}^4\right)\subset L^2(\mathbb{Z}_p^3)\otimes\mathbb{C}^4 \tag{9.1}$$

is satisfied for every t. Here we identify a function of $L^2(\mathbb{Z}_p^3)$ with a function of $L^2(\mathbb{Q}_p^3)$ supported on $\mathbb{Z}_p^3$. A sufficient condition for (9.1) is the following assumption, which we impose from now on (see Lemma 9.2).

Assumption 9.1. *For $l=1,2,3$, the kernel J_l is supported in the unit ball, i.e., $J_l(|y|_p)=0$ for $|y|_p>1$, and $J_l(|y|_p)\in L^1(\mathbb{Z}_p)$.*

For a positive integer $n\geq 1$, take $G_n:=\mathbb{Z}_p/p^n\mathbb{Z}_p$, and use $i=i_0+i_1p+\ldots+i_{n-1}p^{n-1}$, with the $i_k\in\{0,1,\ldots,p-1\}$, as a set of representatives of the elements of the quotient group G_n. We set $G_n^3=(\mathbb{Z}_p/p^n\mathbb{Z}_p)^3=\mathbb{Z}_p^3/p^n\mathbb{Z}_p^3$. Notice that the group G_n^3 has p^{3n} elements. We use $\boldsymbol{i}=(i_1,i_2,i_3)$ to denote the elements of G_n^3, and fix the partition $\mathbb{Z}_p^3=\bigsqcup_{\boldsymbol{i}\in G_n^3}(\boldsymbol{i}+p^n\mathbb{Z}_p^3)$. Taking $\mathfrak{H}_{\mathbb{Z}_p}:=L^2(\mathbb{Z}_p^3)\otimes\mathbb{C}^4$, we have that $\mathfrak{H}_{\mathbb{Z}_p}$ is a closed subspace of $\mathfrak{H}$.

Lemma 9.1. *Set, as in (8.2),*

$$\psi_{\boldsymbol{j},b}(x) = e_b p^{\frac{3n}{2}} \Omega(p^n \|x - \boldsymbol{j}\|_p), \qquad b = 1,2,3,4,\ \boldsymbol{j} \in G_n^3. \tag{9.2}$$

Then $\{\psi_{\boldsymbol{j},b};\ b = 1,2,3,4,\ \boldsymbol{j} \in G_n^3\}$ is an orthonormal set in $\mathfrak{H}$.

Proof. The result follows from $(e_b p^{\frac{3n}{2}} \Omega(p^n \|x - \boldsymbol{j}\|_p),\ e_a p^{\frac{3n}{2}} \Omega(p^n \|x - \boldsymbol{i}\|_p)) = \delta_{b,a}\delta_{\boldsymbol{i},\boldsymbol{j}}$. □

We define the $\mathbb{C}$-vector space $\mathcal{D}_n(\mathbb{Z}_p^3)$ as the space spanned by $\{\Omega(p^n \|x - \boldsymbol{j}\|_p)\}_{\boldsymbol{j} \in G_n^3}$, i.e., $\mathcal{D}_n(\mathbb{Z}_p^3) = V_n \cap L^2(\mathbb{Z}_p^3)$. Notice that the cardinality of this set is p^{3n}.

Lemma 9.2. *Under Assumption 9.1, condition (9.1) holds, and*

$$e^{-\mathrm{i}t\boldsymbol{H}_0}\left(\mathcal{D}_n(\mathbb{Z}_p^3) \otimes \mathbb{C}^4\right) \subset \mathcal{D}_n(\mathbb{Z}_p^3) \otimes \mathbb{C}^4 \subset L^2(\mathbb{Z}_p^3) \otimes \mathbb{C}^4, \qquad t \in \mathbb{R}.$$

Moreover, for $\phi \in \mathcal{D}_n(\mathbb{Z}_p^3)$, formula (8.4) holds with the sum restricted to $k \in G_n$, $k \neq i_l$; the corresponding weighted degree is $d_l = \int_{\mathbb{Z}_p \smallsetminus p^n\mathbb{Z}_p} J_l(|y|_p)dy = \sum_{k \in G_n, k \neq i} w_l(i,k)$.

Proof. Let $\phi \in L^2(\mathbb{Q}_p^3)$ be supported in $\mathbb{Z}_p^3$. If $x_l \notin \mathbb{Z}_p$ and $y \in \mathbb{Z}_p$, then $|x_l - y|_p = |x_l|_p > 1$, so $J_l(|x_l - y|_p) = 0$; hence, by (4.6), $\boldsymbol{J}_{x_l}\phi(x) = 0$ whenever $x_l \notin \mathbb{Z}_p$, and $\boldsymbol{J}_{x_l}\phi(x)$ vanishes if some other coordinate $x_k \notin \mathbb{Z}_p$ because then $\phi(x) = \phi(x^{(l)}(y)) = 0$. Thus $\boldsymbol{J}_{x_l}$ preserves $L^2(\mathbb{Z}_p^3)$, so $\boldsymbol{H}_0$ preserves $\mathfrak{H}_{\mathbb{Z}_p}$, and so does $e^{-\mathrm{i}t\boldsymbol{H}_0}$ (power series). Since also V_n is invariant (Lemma 8.1), the intersection $\mathcal{D}_n(\mathbb{Z}_p^3) \otimes \mathbb{C}^4$ is invariant. In (8.4), the terms with $k \notin G_n$ (i.e., $k \notin \mathbb{Z}_p$) vanish because $J_l(|i-k|_p) = 0$ for $|i-k|_p > 1$. This is consistent with [13, Lemma 13.1]: if $J(|x|_p)$ is an integrable function supported in the unit ball, then

$$\begin{aligned} J(|x|_p) * \Omega(p^n |x-k|_p) = p^{-n} \sum_{i \in G_n \smallsetminus \{k\}} & J(|i-k|_p)\Omega(p^n |x-i|_p) \\ & + \left(\int_{p^n\mathbb{Z}_p} J(|y|_p)dy\right) \Omega(p^n |x-k|_p), \end{aligned}$$

for $k \in G_n$. □

Theorem 9.1. *Under Assumption 9.1, let $\mathcal{I}_n : \ell^2(G_n^3) \otimes \mathbb{C}^4 \to \mathcal{D}_n(\mathbb{Z}_p^3) \otimes \mathbb{C}^4$ be the unitary map sending $\delta_{\boldsymbol{j}} \otimes e_b$ to $\psi_{\boldsymbol{j},b}$. Then $\mathcal{I}_n^{-1}\boldsymbol{H}_0\mathcal{I}_n = \boldsymbol{H}_n$, where $\boldsymbol{H}_n$ is the Hermitian $4p^{3n} \times 4p^{3n}$ matrix given by (8.5), with $\widetilde{G}_n$ replaced by G_n and with the weights and degrees of Lemma 9.2; and $\mathcal{I}_n^{-1}e^{-\mathrm{i}t\boldsymbol{H}_0}\mathcal{I}_n = e^{-\mathrm{i}t\boldsymbol{H}_n}$ for every $t \in \mathbb{R}$. In other words, the restriction of the free p-adic Dirac evolution to $\mathcal{D}_n(\mathbb{Z}_p^3) \otimes \mathbb{C}^4$ is the CTQW on the finite weighted graph G_n^3, with internal space $\mathbb{C}^4$, generated by $\boldsymbol{H}_n$.*

Proof. This follows from Lemmas 8.1 and 9.2, exactly as in the proof of Theorem 8.1. □

Remark 9.1. *Since the dimension of the $\mathbb{C}$-vector space $\mathcal{D}_n(\mathbb{Z}_p^3) \otimes \mathbb{C}^4$ is $4p^{3n}$, the restriction of the operator $\boldsymbol{H}_0$ to $\mathcal{D}_n(\mathbb{Z}_p^3) \otimes \mathbb{C}^4$ is given by a Hermitian matrix of size $4p^{3n} \times 4p^{3n}$, namely the matrix $\boldsymbol{H}_n$ of (8.5), with $\widetilde{G}_n$ replaced by G_n and with the weights of Lemma 9.2. Explicitly, in the basis (9.2),*

$$(\boldsymbol{H}_n)_{(\boldsymbol{i},a),(\boldsymbol{k},b)} = \sum_{l=1}^{3} \left(\boldsymbol{L}^{(l)}\right)_{\boldsymbol{i},\boldsymbol{k}} (\alpha_l)_{ab} + m\,\delta_{\boldsymbol{i},\boldsymbol{k}}\beta_{ab},$$

where $(\boldsymbol{L}^{(l)})_{\boldsymbol{i},\boldsymbol{k}} = w_l(i_l,k_l)$ *if* $\boldsymbol{i}$ *and* $\boldsymbol{k}$ *differ only in the* l*-th coordinate,* $(\boldsymbol{L}^{(l)})_{\boldsymbol{i},\boldsymbol{i}} = -d_l$*, and* $(\boldsymbol{L}^{(l)})_{\boldsymbol{i},\boldsymbol{k}} = 0$ *otherwise. This restriction is the exact discretization of* $\boldsymbol{H}_0$ *on* G_n^3*. In addition, the restriction of* $e^{-\mathrm{it}\boldsymbol{H}_0}$ *to* $\mathcal{D}_n(\mathbb{Z}_p^3)\otimes\mathbb{C}^4$ *is given by the unitary matrix* $e^{-\mathrm{it}\boldsymbol{H}_n}$ *of size* $4p^{3n}\times 4p^{3n}$*.*

9.2. **CTQWs II.** Take $\psi_{\boldsymbol{j},b}(x)$ as in (9.2), and set

$$\Psi_{\boldsymbol{j},b}(t,x) = e^{-\mathrm{it}\boldsymbol{H}_0}\psi_{\boldsymbol{j},b}(x) = \left[\Psi_{\boldsymbol{j},b}^{(1)}(t,x),\ldots,\Psi_{\boldsymbol{j},b}^{(4)}(t,x)\right]^T \in \mathcal{D}_n(\mathbb{Z}_p^3)\otimes\mathbb{C}^4, \qquad t\in\mathbb{R}.$$

Then

$$\|\Psi_{\boldsymbol{j},b}(t,\cdot)\| = 1 \quad \text{for every } t\in\mathbb{R}. \tag{9.3}$$

Now, by Lemma 9.2, $\Psi_{\boldsymbol{j},b}(t,\cdot)\in\mathcal{D}_n(\mathbb{Z}_p^3)\otimes\mathbb{C}^4$, and by Lemma 9.1,

$$\Psi_{\boldsymbol{j},b}(t,x) = \sum_{a=1}^{4}\sum_{\boldsymbol{i}\in G_n^3} c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t)\, e_a p^{\frac{3n}{2}}\Omega(p^n\left\|x-\boldsymbol{i}\right\|_p),$$

where

$$c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t) = (\Psi_{\boldsymbol{j},b}(t,\cdot),\psi_{\boldsymbol{i},a}) = \left(e^{-\mathrm{it}\boldsymbol{H}_n}\right)_{(\boldsymbol{i},a),(\boldsymbol{j},b)}.$$

Remark 9.2. *The condition* $\Omega(p^n\left\|x-\boldsymbol{i}\right\|_p)\Omega(p^n\left\|x-\boldsymbol{j}\right\|_p) = 0$*, for* $\boldsymbol{i}\neq\boldsymbol{j}$*, which follows from the fact that the balls* $\boldsymbol{i}+p^n\mathbb{Z}_p^3$*,* $\boldsymbol{j}+p^n\mathbb{Z}_p^3$ *are disjoint, implies that*

$$\left|\sum_{\boldsymbol{i}\in G_n^3} c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}\, p^{\frac{3n}{2}}\Omega(p^n\left\|x-\boldsymbol{i}\right\|_p)\right|^2 = \sum_{\boldsymbol{i}\in G_n^3}\left|c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}\right|^2 p^{3n}\Omega(p^n\left\|x-\boldsymbol{i}\right\|_p).$$

We now define the transition probabilities. By the Born rule, $\left|c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t)\right|^2$ is the probability of finding the system, at time t, in the state $\psi_{\boldsymbol{i},a}$ (vertex $\boldsymbol{i}$, internal state e_a), given that it was prepared at time 0 in the state $\psi_{\boldsymbol{j},b}$. Since $\Psi_{\boldsymbol{j},b}(t,\cdot)\in\mathcal{D}_n(\mathbb{Z}_p^3)\otimes\mathbb{C}^4$, the probability of finding the particle at the vertex $\boldsymbol{i}$, irrespective of its internal state, is

$$\pi_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t) := \sum_{a=1}^{4}\left|c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t)\right|^2 = \|P_{\boldsymbol{i}}\Psi_{\boldsymbol{j},b}(t,\cdot)\|^2$$

(the two expressions coincide by Theorem 8.1 and Remark 9.2). If the internal state at time 0 is not known, i.e., if the initial state is the mixed state $\rho_{\boldsymbol{j}} := \frac{1}{4}\sum_{b=1}^{4}\psi_{\boldsymbol{j},b}\otimes\overline{\psi_{\boldsymbol{j},b}}$ (the density operator $\frac{1}{4}\sum_b|\psi_{\boldsymbol{j},b}\rangle\langle\psi_{\boldsymbol{j},b}|$), then the probability of finding the particle at $\boldsymbol{i}$ at time t is

$$\pi_{\boldsymbol{i},\boldsymbol{j}}(t) := \mathrm{Tr}\left(P_{\boldsymbol{i}}\, e^{-\mathrm{it}\boldsymbol{H}_0}\rho_{\boldsymbol{j}}e^{\mathrm{it}\boldsymbol{H}_0}\right) = \frac{1}{4}\sum_{b=1}^{4}\pi_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t) = \frac{1}{4}\sum_{a,b=1}^{4}\left|c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t)\right|^2.$$

Then, from (9.3) and Remark 9.2, we obtain that

$$1 = \sum_{a=1}^{4}\left\|\sum_{\boldsymbol{i}\in G_n^3} c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}p^{\frac{3n}{2}}\Omega(p^n\left\|x-\boldsymbol{i}\right\|_p)\right\|_2^2 = \sum_{a=1}^{4}\sum_{\boldsymbol{i}\in G_n^3}\left|c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}\right|^2 = \sum_{\boldsymbol{i}\in G_n^3}\pi_{\boldsymbol{i},\boldsymbol{j}}^{(b)}(t).$$

We have thus established the following result.

Theorem 9.2. *Under Assumption 9.1, for every $t \in \mathbb{R}$:*

(i) for every $b \in \{1,2,3,4\}$, the matrix $[\pi^{(b)}_{\boldsymbol{i},\boldsymbol{j}}(t)]_{\boldsymbol{i},\boldsymbol{j}\in G_n^3}$ is stochastic, i.e., $\pi^{(b)}_{\boldsymbol{i},\boldsymbol{j}}(t) \in [0,1]$ and, for any $\boldsymbol{j} \in G_n^3$, $\sum_{\boldsymbol{i}\in G_n^3} \pi^{(b)}_{\boldsymbol{i},\boldsymbol{j}}(t) = 1$;

(ii) the matrix $[\pi_{\boldsymbol{i},\boldsymbol{j}}(t)]_{\boldsymbol{i},\boldsymbol{j}\in G_n^3}$ is doubly stochastic, i.e., $\sum_{\boldsymbol{i}} \pi_{\boldsymbol{i},\boldsymbol{j}}(t) = 1$ for every $\boldsymbol{j}$, and $\sum_{\boldsymbol{j}} \pi_{\boldsymbol{i},\boldsymbol{j}}(t) = 1$ for every $\boldsymbol{i}$.

Consequently, for each fixed t, $\pi_{\boldsymbol{i},\boldsymbol{j}}(t)$ is the one-step transition kernel of a random walk on the state space G_n^3, and the uniform distribution on G_n^3 is invariant under it.

Proof. Part (i) was established above. For (ii), the column sums equal 1 by (i). For the row sums, let $U = e^{-\mathrm{i}t\boldsymbol{H}_n}$, a unitary matrix with entries $U_{(\boldsymbol{i},a),(\boldsymbol{j},b)} = c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t)$. Each row of a unitary matrix has norm one, so $\sum_{\boldsymbol{j}} \pi_{\boldsymbol{i},\boldsymbol{j}}(t) = \frac{1}{4}\sum_{a=1}^{4}\sum_{\boldsymbol{j},b} \left|U_{(\boldsymbol{i},a),(\boldsymbol{j},b)}\right|^2 = \frac{1}{4}\cdot 4 = 1$. □

The family $\{[\pi_{\boldsymbol{i},\boldsymbol{j}}(t)]\}_{t\geq 0}$ is *not*, however, the family of transition matrices of a continuous-time Markov chain. This is a general feature of CTQWs, see [39, 42], and it is the precise sense in which a CTQW differs from a classical random walk:

Proposition 9.1. *Assume that $\boldsymbol{H}_n$ has a non-zero matrix entry between two different vertices, i.e., $(\boldsymbol{H}_n)_{(\boldsymbol{i},a),(\boldsymbol{k},b)} \neq 0$ for some $\boldsymbol{i} \neq \boldsymbol{k}$ and some a, b (this is the case as soon as some $w_l(i,k) \neq 0$). Then there is no matrix-valued function $\Pi(t)$ satisfying $\Pi(t+s) = \Pi(t)\Pi(s)$ for all $t, s \geq 0$ and $\Pi(t) = [\pi_{\boldsymbol{i},\boldsymbol{j}}(t)]$ for all $t \geq 0$; the same holds for $[\pi^{(b)}_{\boldsymbol{i},\boldsymbol{j}}(t)]$, $b = 1,2,3,4$.*

Proof. Write $\pi(t) = [\pi_{\boldsymbol{i},\boldsymbol{j}}(t)]$. Since $e^{-\mathrm{i}t\boldsymbol{H}_n} = \mathrm{Id} - \mathrm{i}t\boldsymbol{H}_n + O(t^2)$, for $\boldsymbol{i} \neq \boldsymbol{j}$ we have $c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t) = -\mathrm{i}t(\boldsymbol{H}_n)_{(\boldsymbol{i},a),(\boldsymbol{j},b)} + O(t^2)$, so $\pi_{\boldsymbol{i},\boldsymbol{j}}(t) = \frac{t^2}{4}\sum_{a,b}\left|(\boldsymbol{H}_n)_{(\boldsymbol{i},a),(\boldsymbol{j},b)}\right|^2 + O(t^3)$, and $\pi(t) = \mathrm{Id} + O(t^2)$ (the diagonal entries are determined by the column sums). Suppose $\pi(t+s) = \pi(t)\pi(s)$ for $t, s \geq 0$. Then π is a continuous semigroup of finite matrices with $\pi(0) = \mathrm{Id}$, so $\pi(t) = e^{tQ}$ with $Q = \pi'(0) = 0$; hence $\pi(t) = \mathrm{Id}$ for all $t \geq 0$. But, for the pair $\boldsymbol{i} \neq \boldsymbol{k}$ in the hypothesis, $\pi_{\boldsymbol{i},\boldsymbol{k}}(t) \geq \frac{t^2}{4}\left|(\boldsymbol{H}_n)_{(\boldsymbol{i},a),(\boldsymbol{k},b)}\right|^2 + O(t^3) > 0$ for small $t > 0$, a contradiction. The same argument applies to $\pi^{(b)}(t)$, with b as in the hypothesis. □

Thus, tracing out the internal (spinor) degrees of freedom of the p-adic Dirac CTQW produces, at every fixed time t, a stochastic (and, after averaging over the initial internal state, doubly stochastic) matrix on the finite graph G_n^3; the time evolution of these matrices is quantum, not Markovian.

Remark 9.3. *By Parseval's identity, the amplitudes can be computed in the Fourier representation: $c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t) = \langle \widehat{\Psi}^{(a)}_{\boldsymbol{j},b}(t,\xi), p^{\frac{-3n}{2}}\chi_p(\boldsymbol{i}\cdot\xi)\Omega(\|p^n\xi\|_p)\rangle$, since the Fourier transform of $p^{\frac{3n}{2}}\Omega(p^n\|x-\boldsymbol{i}\|_p)$ is $p^{\frac{-3n}{2}}\chi_p(\boldsymbol{i}\cdot\xi)\Omega(\|p^n\xi\|_p)$.*

9.3. **Computation of $\Psi_{\boldsymbol{j},b}(t,x)$.** Consider the Cauchy problem

$$\begin{cases} \mathrm{i}\frac{\partial}{\partial t}\Psi_{\boldsymbol{j},b}(t,x) = \mathcal{F}^{-1}_{\xi\to x}\left\{h(\xi)\mathcal{F}_{x\to\xi}\left(\Psi_{\boldsymbol{j},b}(t,x)\right)\right\}, \quad x \in \mathbb{Q}_p^3,\ t \in \mathbb{R}, \\ \Psi_{\boldsymbol{j},b}(0,x) = \psi_{\boldsymbol{j},b}(x) \in \mathfrak{H}_{\mathbb{Z}_p}. \end{cases} \tag{9.4}$$

By passing to the Fourier transform in (9.4),

$$\begin{cases} \mathrm{i}\frac{\partial}{\partial t}\widehat{\Psi_{\boldsymbol{j},b}}(t,\xi)=h(\xi)\widehat{\Psi_{\boldsymbol{j},b}}(t,\xi), \quad \xi\in\mathbb{Q}_p^3,\ t\in\mathbb{R},\\ \widehat{\Psi_{\boldsymbol{j},b}}(0,\xi)=\widehat{\psi_{\boldsymbol{j},b}}(\xi), \end{cases}$$

and using $h(\xi)=u(\xi)^{-1}\beta\lambda(\xi)u(\xi)$, cf. (6.3), we have

$$\Psi_{\boldsymbol{j},b}(t,x)=\mathcal{F}_{\xi\to x}^{-1}\left\{u(\xi)^{-1}e^{-\mathrm{i}t\beta\lambda(\xi)}u(\xi)\widehat{\psi_{\boldsymbol{j},b}}(\xi)\right\}\in\mathcal{D}_n(\mathbb{Z}_p^3)\otimes\mathbb{C}^4,\qquad t\in\mathbb{R}.$$

Since $h(\xi)^2=\lambda(\xi)^2 I_4$, the matrix exponential can be written without diagonalizing:

$$u(\xi)^{-1}e^{-\mathrm{i}t\beta\lambda(\xi)}u(\xi)=e^{-\mathrm{i}th(\xi)}=\cos(t\lambda(\xi))I_4-\mathrm{i}\,\frac{\sin(t\lambda(\xi))}{\lambda(\xi)}h(\xi),$$

with the convention $\sin(t\lambda)/\lambda=t$ if $\lambda=0$. Using $\widehat{\psi_{\boldsymbol{j},b}}(\xi)=p^{-3n/2}\chi_p(\boldsymbol{j}\cdot\xi)\Omega(\|p^n\xi\|_p)e_b$, we obtain the explicit formula for the amplitudes

$$c_{(\boldsymbol{i},a),(\boldsymbol{j},b)}(t)=p^{-3n}\int_{\|\xi\|_p\leq p^n}\chi_p((\boldsymbol{j}-\boldsymbol{i})\cdot\xi)\left\{\cos(t\lambda(\xi))\delta_{ab}-\mathrm{i}\frac{\sin(t\lambda(\xi))}{\lambda(\xi)}h_{ab}(\xi)\right\}d^3\xi.$$

Under Assumption 9.1, $\mathbf{j}(\xi)$ vanishes on $\mathbb{Z}_p^3$ and takes finitely many values on $\{\|\xi\|_p\leq p^n\}$ (it depends only on $|\xi_l|_p\in\{p^0,\dots,p^n\}$), so the integral is a finite sum of integrals of additive characters over products of p-adic spheres, which can be evaluated in closed form.

10. Towards quantum networks based on the Dirac equation

The constructions of Sections 8 and 9 were carried out for their own sake, as exact discretizations of the free p-adic Dirac equation into CTQWs. We now argue that these results suggest the feasibility of quantum networks whose elementary dynamics are Dirac-, rather than Schrödinger-, type, and discuss how they relate to the broader state of the art reviewed in Section 2.

10.1. **From a single walk to a quantum network.** Theorem 9.2 shows that the matrix $[\pi_{\boldsymbol{i},\boldsymbol{j}}(t)]_{\boldsymbol{i},\boldsymbol{j}\in G_n^3}$, obtained by summing the Born-rule weights of the four spinor components at each node and averaging over the initial internal state, is a doubly stochastic matrix for every t. In other words, tracing out the internal degrees of freedom of the p-adic Dirac CTQW produces, at every fixed time, the transition kernel of a random walk on the finite graph G_n^3, while the time dependence remains genuinely quantum (Proposition 9.1). This is the kind of object used as the substrate for the quantum neural network introduced in [17]. In that paper, the authors present a new class of quantum neural networks (QNNs) whose states are solutions of p-adic Schrödinger equations with a non-local potential that controls the interaction between the neurons. These equations are obtained as Wick rotations of the state equations of p-adic cellular neural networks (CNNs), [52–54]. p-Adic CNNs arise as continuous limits of large discrete hierarchical neural networks (NNs). They are bio-inspired by the Wilson–Cowan model, which describes the macroscopic dynamics of large neuronal populations. Here, we propose a similar construction using the p-adic Dirac equations introduced here.

First, we introduce a kernel matrix

$$W(x,y)=[W_{ab}(x,y)]_{a,b=1}^{4}\,,\qquad x,y\in\mathbb{Q}_p^3,$$

where $W_{ab}(x,y)\in\mathbb{C}$. The entry $W_{ab}(x,y)$ describes the strength of the connection between the b-th spinor component at position y and the a-th spinor component at position x. We fix an activation function $\phi:\mathbb{R}\to\mathbb{R}$; for instance $\phi(s)=\tanh(s)$ or $\phi(s)=\frac{1}{2}\,|s+1|-\frac{1}{2}\,|s-1|$. We extend ϕ to the complex numbers by taking $\phi(a+\mathrm{i}b)=\phi(a)+\mathrm{i}\phi(b)$, and to $\mathbb{C}^4$ componentwise:

$$\phi\left(\Psi(t,y)\right)=\phi\left(\begin{bmatrix}\Psi^{(1)}(t,y)\\ \vdots\\ \Psi^{(4)}(t,y)\end{bmatrix}\right)=\begin{bmatrix}\phi(\operatorname{Re}\Psi^{(1)}(t,y))+\mathrm{i}\phi(\operatorname{Im}\Psi^{(1)}(t,y))\\ \vdots\\ \phi(\operatorname{Re}\Psi^{(4)}(t,y))+\mathrm{i}\phi(\operatorname{Im}\Psi^{(4)}(t,y))\end{bmatrix}\in\mathbb{C}^4,$$

and we fix a bias $z(t,x)=[z^{(1)}(t,x),\dots,z^{(4)}(t,x)]^T$. With this notation, we introduce the Dirac analogue of the QNN introduced in [17]:

$$\mathrm{i}\frac{\partial}{\partial t}\Psi(t,x)=\boldsymbol{H}_0\Psi(t,x)+\int_{\mathbb{Q}_p^3}W(x,y)\,\phi\left(\Psi(t,y)\right)d^3y+z(t,x),\qquad t\geq 0,\ x\in\mathbb{Q}_p^3,\tag{10.1}$$

where $W(x,y)\phi(\Psi(t,y))$ denotes the product of a 4×4 matrix and a vector of $\mathbb{C}^4$. We make the following assumptions:

(H1) $W_{ab}\in L^2(\mathbb{Q}_p^3\times\mathbb{Q}_p^3)$ for $a,b=1,\dots,4$;
(H2) $\phi:\mathbb{R}\to\mathbb{R}$ is Lipschitz continuous, with Lipschitz constant L_ϕ, and $\phi(0)=0$;
(H3) $z\in C([0,\infty),\mathfrak{H})$.

Both activation functions mentioned above satisfy (H2) with $L_\phi=1$. The condition $\phi(0)=0$ cannot be dropped: since $\mathbb{Q}_p^3$ has infinite measure, a non-zero constant does not belong to $L^2(\mathbb{Q}_p^3)$.

Lemma 10.1. *Assume (H1)–(H2). Then:*

(i) the integral operator $(\mathcal{W}\Phi)(x):=\int_{\mathbb{Q}_p^3}W(x,y)\Phi(y)d^3y$ is a bounded (Hilbert–Schmidt) operator on $\mathfrak{H}$, with $\|\mathcal{W}\|\leq\|W\|_2:=(\sum_{a,b}\|W_{ab}\|^2_{L^2(\mathbb{Q}_p^3\times\mathbb{Q}_p^3)})^{1/2}$;

(ii) the Nemytskii (superposition) operator $\Phi(\Psi)(y):=\phi(\Psi(y))$ maps $\mathfrak{H}$ into $\mathfrak{H}$, and $\|\Phi(\Psi)-\Phi(\Psi')\|\leq L_\phi\,\|\Psi-\Psi'\|$ for all $\Psi,\Psi'\in\mathfrak{H}$.

Proof. (i) By the Cauchy–Schwarz inequality, $|(\mathcal{W}\Phi)_a(x)|^2\leq\left(\sum_b\int|W_{ab}(x,y)|^2\,d^3y\right)\|\Phi\|^2$; integrate in x and sum over a. (ii) For $a,b,a',b'\in\mathbb{R}$, $|\phi(a)+\mathrm{i}\phi(b)-\phi(a')-\mathrm{i}\phi(b')|^2\leq L_\phi^2(|a-a'|^2+|b-b'|^2)$. Applying this componentwise and pointwise, and integrating, gives the Lipschitz estimate; taking $\Psi'=0$ and using $\phi(0)=0$ gives $\|\Phi(\Psi)\|\leq L_\phi\,\|\Psi\|$, so $\Phi(\mathfrak{H})\subset\mathfrak{H}$. See [48] for the general theory of superposition operators. □

Proposition 10.1. *Assume (H1)–(H3). Then, for every $\Psi_0\in\mathfrak{H}$, equation (10.1) has a unique mild solution $\Psi\in C([0,\infty),\mathfrak{H})$ with $\Psi(0)=\Psi_0$, i.e., a unique continuous solution of*

$$\Psi(t)=e^{-\mathrm{i}t\boldsymbol{H}_0}\Psi_0-\mathrm{i}\int_0^t e^{-\mathrm{i}(t-s)\boldsymbol{H}_0}\left\{\mathcal{W}\Phi(\Psi(s))+z(s)\right\}ds.$$

Moreover, $\Psi \in C^1([0,\infty),\mathfrak{H})$ and (10.1) holds in $\mathfrak{H}$ for every $t \geq 0$.

Proof. By Lemma 10.1, the map $F(t,\Psi) := -\mathrm{i}\{\mathcal{W}\Phi(\Psi)+z(t)\}$ is continuous in t and globally Lipschitz in Ψ, with constant $\|W\|_2 L_\phi$, uniformly in t. Since $e^{-\mathrm{i}t\boldsymbol{H}_0}$ is a uniformly continuous unitary group, the standard Picard iteration (equivalently, the Banach fixed-point theorem in $C([0,T],\mathfrak{H})$ with an exponentially weighted norm) gives a unique global mild solution, see [47, Theorem 6.1.2]; since $\boldsymbol{H}_0$ is bounded and F is continuous, the mild solution is a classical C^1 solution. □

In a discretization of (10.1) on G_n^3, i.e., restricting to $\mathcal{D}_n(\mathbb{Z}_p^3)\otimes\mathbb{C}^4$ and replacing W by a $4p^{3n}\times 4p^{3n}$ matrix, (10.1) becomes a system of $4p^{3n}$ ordinary differential equations with globally Lipschitz right-hand side, which again has unique global solutions.

Remark 10.1. *For $W \neq 0$ or $z \neq 0$, the evolution (10.1) is nonlinear and, in general, it does not preserve the norm: for a solution Ψ, using that $(\boldsymbol{H}_0\Psi,\Psi)$ is real,*

$$\frac{d}{dt}\|\Psi(t)\|^2 = 2\,\mathrm{Im}\,(\mathcal{W}\Phi(\Psi(t))+z(t),\Psi(t))\,,$$

which is in general non-zero. Hence, for $W \neq 0$, the network dynamics is not unitary, and $\|\Psi(t)\|^2$ cannot, in general, be interpreted as a total probability without renormalization. We stress that (10.1) is a nonlinear equation for a vector state, and therefore it is not *of Lindblad (Gorini–Kossakowski–Sudarshan–Lindblad) type, since GKSL equations are linear equations for density operators [49–51]. Whether the dynamics (10.1) can be obtained as the reduced dynamics of a larger, closed or open, quantum system—as is suggested by the interpretation of $W\phi(\Psi)$ as an effective interaction with an environment—is an open problem.*

In such a discretization, the vertices $\boldsymbol{j} \in G_n^3$ play the role of neurons or network nodes, arranged, as in [17], in the hierarchical, ultrametric tree structure inherited from $\mathbb{Z}_p^3$, while the four-dimensional fiber $\mathbb{C}^4$ attached to each node carries four-component spinor degrees of freedom that are absent from the scalar p-adic Schrödinger networks studied previously. A Dirac-based quantum network built along these lines would differ from the QNN family studied in [17] in a structurally new way: the free part of the dynamics has the charge-conjugation symmetry of Section 7, which on the graph reads $(\mathrm{Id}\otimes\alpha_2)\overline{\boldsymbol{H}_n}(\mathrm{Id}\otimes\alpha_2)^{-1} = -\boldsymbol{H}_n$ (because the matrices $\boldsymbol{L}^{(l)}$ are real) and relates pairs of internal channels at each node, and the topologically protected zero modes of the p-adic Jackiw–Rebbi model, [14], are natural candidates for noise-resilient network nodes. Adding the non-local weight kernel W and the nonlinear activation ϕ turns the free, unitary Dirac CTQW ($W = 0$) into a nonlinear, non-unitary network dynamics ($W \neq 0$), whose physical interpretation is discussed in Remark 10.1.

10.2. **Relation to the broader state of the art.** As discussed in Section 2, most of the literature connecting quantum walks to the Dirac equation is built on discrete-time, coined walks [18–21, 33–36], with the Dirac equation appearing as a continuum *limit*. Continuous-time dynamics with spinor-like structure on graphs have also been constructed, notably the

Dirac-type CTQWs on periodic lattices of Childs and Goldstone [22], the topological Dirac equation on networks of Bianconi [23], CTQWs on bipartite graphs [24], and honeycomb tight-binding models [25, 26]. The construction developed in this paper differs from these in three respects: (a) it is formulated on hierarchical, ultrametric graphs, the edge weights being determined by the p-adic tree; (b) the walk is not an approximation of the Dirac equation, but the exact restriction of the p-adic Dirac evolution to an invariant subspace (Theorems 8.1 and 9.1); and (c) it is set up as the substrate of a quantum network. To the best of our knowledge, it is the first CTQW with these properties.

11. Discussion and conclusions

11.1. **Summary.** We introduced a new p-adic version of the Dirac equation, formulated in the standard axiomatic (Dirac–von Neumann) framework of quantum mechanics, in which the usual momentum operators are replaced by bounded non-local operators with real radial kernels. Working in momentum space, we constructed its plane-wave solutions (Section 5), diagonalized the free Hamiltonian, and determined its spectrum (Section 6), which is pure point, with eigenvalues of infinite multiplicity; in the massless case we described the zero-energy subspace. We identified the charge-conjugation symmetry of the model (Section 7): it is implemented by $\mathcal{C}\Psi = \alpha_2\overline{\Psi}$, satisfies $\mathcal{C}^2 = -1$, exchanges positive- and negative-energy states of the free Hamiltonian, and, in an external field, reverses both the charge and the vector potential.

Building on this, we discretized the free equation in two ways, both giving CTQWs rather than discrete-time, coined walks. The first discretization is set on a countable graph (Section 8); the second, more explicit one is set on a finite, tree-like graph (Section 9). In both cases the discretization is exact, and the walk Hamiltonian is a Dirac-type combination of weighted graph Laplacians with ultrametric weights. For the second construction we proved that the transition probabilities, once the four internal components of the wavefunction are added together, form a stochastic matrix at every moment in time, which becomes doubly stochastic after averaging over the initial internal state; as for every non-trivial CTQW, the time evolution of these matrices is not Markovian.

We then used these results to argue, in Section 10, that the same construction can serve as the foundation for a quantum network, extending earlier work on p-adic quantum neural networks [17] from scalar (Schrödinger-type) internal states to four-component spinor states, and we proved the global well-posedness of the network equation under Lipschitz assumptions.

11.2. **Open problems.** The results above raise more questions than they close. We list the ones we regard as the most pressing.

(1) **Quantitative continuum limit.** Although each walk of Sections 8 and 9 is an exact restriction of the p-adic Dirac evolution, the initial states that are not constant on balls of radius p^{-n} are only approximated by the walk at level n. We have not established a quantitative estimate, analogous to the convergence rates proved by

Arrighi, Nesme, and Forets for the higher-dimensional Archimedean DTQW–Dirac correspondence [21], for the convergence of $P_{V_n} e^{-\mathrm{i}t\boldsymbol{H}_0} P_{V_n}\psi$ to $e^{-\mathrm{i}t\boldsymbol{H}_0}\psi$ as $n \to \infty$ in terms of the regularity of ψ. Since $\mathbb{Z}_p^3$ is compact and ultrametric, one may expect a rate governed by p^{-n}; making this precise is open.

(2) **Interactions and gauge fields.** The Dirac equation studied here is free. Section 7 shows that, with the standard minimal coupling, charge conjugation also reverses the vector potential. Coupling $\boldsymbol{H}_0$ to a genuinely p-adic electromagnetic (or more general gauge) potential, in the spirit of the gauge-covariant discrete-time walks of Arnault and Debbasch [28] and of [44] in the Archimedean setting, has not been carried out; nor is it known whether the resulting equation would still discretize into a CTQW with a stochastic-matrix structure analogous to the one established in Section 9.

(3) **Momentum-type operators with odd symbols.** The operators $\boldsymbol{J}_{x_l}$ have even (radial) symbols, which is the origin of the differences with the standard charge conjugation (Remark 7.1). Operators of the form $\mathrm{i}K*$, with K real and odd ($K(-y) = -K(y)$), are self-adjoint and anticommute with complex conjugation, like $-\mathrm{i}\partial_x$; such kernels exist on $\mathbb{Q}_p$ (e.g., $K(y) = f(|y|_p)\sin(2\pi\{y\}_p)$), but they are not radial. Whether a p-adic Dirac equation built from such operators has good spectral and discretization properties is open.

(4) **Curved p-adic space-time.** Di Molfetta, Brachet, and Debbasch showed that a family of discrete-time walks converges to massless Dirac fermions on a curved background metric [44]. Whether a meaningful notion of curvature exists for $\mathbb{Q}_p^3$, and whether the CTQW construction of this paper can be adapted to such a background, is entirely open.

(5) **Nonlinear network dynamics.** Proposition 10.1 gives global well-posedness of the network equation (10.1) under Lipschitz assumptions. Its long-time behaviour, the role of the charge-conjugation symmetry when $W \neq 0$, and the question of whether it can be derived as the reduced dynamics of an open quantum system (Remark 10.1) remain open.

(6) **Entanglement structure.** We have not characterized the entanglement generated by the free Dirac CTQW across the hierarchical levels of the p-adic tree G_n^3, nor the entanglement between the four spinor channels at a fixed node. Understanding this structure is a prerequisite for assessing whether the construction offers any genuine advantage, as a quantum network, beyond that already available for scalar CTQWs.

(7) **Topological protection on the walk.** The p-adic Jackiw–Rebbi model of [14] produces topologically protected, localized zero modes for a p-adic Dirac Hamiltonian. We have not constructed the CTQW that discretizes that model, nor determined whether its zero modes survive discretization on G_n^3 as noise-resilient network nodes, as suggested informally in Section 10.

We regard item (5) as the most immediate next step, since it extends the machinery already developed in [17] for the scalar case to the four-spinor setting of the present paper;

items (2), (3), (4), and (7), by contrast, appear to require genuinely new ideas, specific to the interplay between p-adic analysis and the structure of the Dirac equation, and we leave them for future work.

12. Appendix: Basic facts on p-adic analysis

In this section, we fix the notation and collect some basic results on p-adic analysis that we use throughout the article. For a detailed exposition on p-adic analysis, the reader may consult [6, 45, 55, 56].

12.1. **The field of p-adic numbers.** Along this article p denotes a prime number. The field of p-adic numbers $\mathbb{Q}_p$ is defined as the completion of the field of rational numbers $\mathbb{Q}$ with respect to the p-adic norm $|\cdot|_p$, which is defined as

$$|x|_p = \begin{cases} 0 & \text{if } x = 0, \\ p^{-\gamma} & \text{if } x = p^{\gamma}\frac{a}{b}, \end{cases}$$

where a and b are integers coprime with p. The integer $\gamma = ord_p(x) := ord(x)$, with $ord(0) := +\infty$, is called the p-adic order of x. We extend the p-adic norm to $\mathbb{Q}_p^N$ by taking $\|x\|_p := \max_{1\leq i\leq N} |x_i|_p$ for $x = (x_1, \ldots, x_N) \in \mathbb{Q}_p^N$. By defining $ord(x) = \min_{1\leq i\leq N}\{ord(x_i)\}$, we have $\|x\|_p = p^{-ord(x)}$. The metric space $(\mathbb{Q}_p^N, \|\cdot\|_p)$ is a complete ultrametric space. As a topological space $\mathbb{Q}_p$ is homeomorphic to a Cantor-like subset of the real line, see, e.g., [6,45].

Any p-adic number $x \neq 0$ has a unique expansion of the form $x = p^{ord(x)} \sum_{j=0}^{\infty} x_j p^j$, where $x_j \in \{0, 1, 2, \ldots, p-1\}$ and $x_0 \neq 0$. In addition, any $x \in \mathbb{Q}_p^N \setminus \{0\}$ can be represented uniquely as $x = p^{ord(x)} v$, where $\|v\|_p = 1$.

12.2. **Topology of $\mathbb{Q}_p^N$.** For $r \in \mathbb{Z}$, denote by $B_r^N(a) = \{x \in \mathbb{Q}_p^N; \|x - a\|_p \leq p^r\}$ the ball of radius p^r with center at $a = (a_1, \ldots, a_N) \in \mathbb{Q}_p^N$, and take $B_r^N(0) := B_r^N$. Note that $B_r^N(a) = B_r(a_1) \times \cdots \times B_r(a_N)$, where $B_r(a_i) := \{x \in \mathbb{Q}_p; |x_i - a_i|_p \leq p^r\}$ is the one-dimensional ball of radius p^r with center at $a_i \in \mathbb{Q}_p$. The ball B_0^N equals the product of N copies of $B_0 = \mathbb{Z}_p$, the ring of p-adic integers. We also denote by $S_r^N(a) = \{x \in \mathbb{Q}_p^N; \|x - a\|_p = p^r\}$ the sphere of radius p^r with center at a, and take $S_r^N(0) := S_r^N$. In addition, two balls in $\mathbb{Q}_p^N$ are either disjoint or one is contained in the other.

As a topological space $(\mathbb{Q}_p^N, \|\cdot\|_p)$ is totally disconnected, i.e., the only connected subsets of $\mathbb{Q}_p^N$ are the empty set and the points. A subset of $\mathbb{Q}_p^N$ is compact if and only if it is closed and bounded in $\mathbb{Q}_p^N$, see, e.g., [6, Section 1.3], or [45, Section 1.8]. The balls and spheres are compact subsets. Thus $(\mathbb{Q}_p^N, \|\cdot\|_p)$ is a locally compact topological space.

12.3. **The Haar measure.** Since $(\mathbb{Q}_p^N, +)$ is a locally compact topological group, there exists a Haar measure $d^N x$, which is invariant under translations, i.e., $d^N(x+a) = d^N x$, [57]. If we normalize this measure by the condition $\int_{\mathbb{Z}_p^N} d^N x = 1$, then $d^N x$ is unique.

Notation 1. *We will use $\Omega(p^{-r}\|x - a\|_p)$ to denote the characteristic function of the ball $B_r^N(a) = a + p^{-r}\mathbb{Z}_p^N$, where $\mathbb{Z}_p^N = \{x \in \mathbb{Q}_p^N; \|x\|_p \leq 1\}$ is the N-dimensional unit ball. For more general sets, we will use the notation 1_A for the characteristic function of the set A.*

12.4. **The Bruhat–Schwartz space.** A complex-valued function φ defined on $\mathbb{Q}_p^N$ is called locally constant if for any $x \in \mathbb{Q}_p^N$ there exists an integer $l(x) \in \mathbb{Z}$ such that

$$\varphi(x + x') = \varphi(x) \text{ for any } x' \in B_{l(x)}^N. \tag{12.1}$$

A function $\varphi : \mathbb{Q}_p^N \to \mathbb{C}$ is called a Bruhat–Schwartz function (or a test function) if it is locally constant with compact support. Any test function can be represented as a linear combination, with complex coefficients, of characteristic functions of balls. The $\mathbb{C}$-vector space of Bruhat–Schwartz functions is denoted by $\mathcal{D}(\mathbb{Q}_p^N)$. For $\varphi \in \mathcal{D}(\mathbb{Q}_p^N)$, the largest number $l = l(\varphi)$ satisfying (12.1) is called the exponent of local constancy (or the parameter of constancy) of φ.

12.5. L^ρ **spaces.** Given $\rho \in [1, \infty)$, we denote by $L^\rho(\mathbb{Q}_p^N) := L^\rho(\mathbb{Q}_p^N, d^N x)$ the $\mathbb{C}$-vector space of all the complex valued functions g satisfying $\|g\|_\rho = \left(\int_{\mathbb{Q}_p^N} |g(x)|^\rho \, d^N x\right)^{1/\rho} < \infty$, where $d^N x$ is the normalized Haar measure on $(\mathbb{Q}_p^N, +)$. For $\rho = \infty$, $L^\infty(\mathbb{Q}_p^N)$ denotes the space of essentially bounded measurable functions, with the norm $\|g\|_\infty = \operatorname{ess\,sup}_{x \in \mathbb{Q}_p^N} |g(x)|$. If U is an open subset of $\mathbb{Q}_p^N$, $\mathcal{D}(U)$ denotes the $\mathbb{C}$-vector space of test functions with supports contained in U; then $\mathcal{D}(U)$ is dense in $L^\rho(U) = \{\varphi : U \to \mathbb{C}; \|\varphi\|_\rho = \{\int_U |\varphi(x)|^\rho \, d^N x\}^{1/\rho} < \infty\}$, for $1 \leq \rho < \infty$, see, e.g., [45, Section 4.3]. We denote by $L_{\mathbb{R}}^\rho(U)$ the real counterpart of $L^\rho(U)$.

12.6. **The Fourier transform.** We recall that a p-adic number $x \neq 0$ has a unique expansion of the form $x = p^{ord(x)} \sum_{j=0}^{\infty} x_j p^j$, where $x_j \in \{0, \ldots, p-1\}$ and $x_0 \neq 0$. By using this expansion, we define the fractional part of $x \in \mathbb{Q}_p$, denoted $\{x\}_p$, as the rational number

$$\{x\}_p = \begin{cases} 0 & \text{if } x = 0 \text{ or } ord(x) \geq 0, \\ p^{ord(x)} \sum_{j=0}^{-ord(x)-1} x_j p^j & \text{if } ord(x) < 0. \end{cases}$$

Set $\chi_p(y) = \exp(2\pi \mathrm{i} \{y\}_p)$ for $y \in \mathbb{Q}_p$. The map $\chi_p(\cdot)$ is an additive character on $\mathbb{Q}_p$, i.e., a continuous map from $(\mathbb{Q}_p, +)$ into S (the unit circle considered as a multiplicative group) satisfying $\chi_p(x_0 + x_1) = \chi_p(x_0)\chi_p(x_1)$, $x_0, x_1 \in \mathbb{Q}_p$. It satisfies $|\chi_p(y)| = 1$, and $\chi_p(y) = 1$ for $y \in \mathbb{Z}_p$; in particular, χ_p is bounded, but it does not belong to $L^2(\mathbb{Q}_p)$. The additive characters of $\mathbb{Q}_p$ form an Abelian group which is isomorphic to $(\mathbb{Q}_p, +)$. The isomorphism is given by $\xi \mapsto \chi_p(\xi x)$, see, e.g., [45, Section 2.3].

Set $\xi \cdot x := \sum_{j=1}^N \xi_j x_j$, for $\xi = (\xi_1, \ldots, \xi_N)$, $x = (x_1, \ldots, x_N) \in \mathbb{Q}_p^N$, as before. The Fourier transform of $\varphi \in \mathcal{D}(\mathbb{Q}_p^N)$ is defined as

$$\mathcal{F}\varphi(\xi) = \int_{\mathbb{Q}_p^N} \chi_p(\xi \cdot x)\varphi(x) d^N x \quad \text{for } \xi \in \mathbb{Q}_p^N,$$

where $d^N x$ is the normalized Haar measure on $\mathbb{Q}_p^N$. The Fourier transform is a linear isomorphism from $\mathcal{D}(\mathbb{Q}_p^N)$ onto itself satisfying

$$(\mathcal{F}(\mathcal{F}\varphi))(\xi) = \varphi(-\xi), \tag{12.2}$$

see, e.g., [45, Section 4.8]. We also use the notation $\mathcal{F}_{x\to\xi}\varphi$ and $\widehat{\varphi}$ for the Fourier transform of φ. For instance, $\mathcal{F}(\Omega(p^{-r}\left\|x-a\right\|_p))(\xi)=p^{rN}\chi_p(a\cdot\xi)\Omega(p^r\left\|\xi\right\|_p)$.

The Fourier transform extends to $L^2(\mathbb{Q}_p^N)$. If $f\in L^2(\mathbb{Q}_p^N)$, its Fourier transform is defined as

$$(\mathcal{F}f)(\xi)=\lim_{k\to\infty}\int_{\left\|x\right\|_p\leq p^k}\chi_p(\xi\cdot x)f(x)d^Nx,\quad \text{for } \xi\in\mathbb{Q}_p^N,$$

where the limit is taken in $L^2(\mathbb{Q}_p^N)$. We recall that the Fourier transform is unitary on $L^2(\mathbb{Q}_p^N)$, i.e., $\left\|f\right\|_2=\left\|\mathcal{F}f\right\|_2$ for $f\in L^2(\mathbb{Q}_p^N)$, and that (12.2) is also valid in $L^2(\mathbb{Q}_p^N)$, see, e.g., [56, Chapter III, Section 2]. Moreover, for $J\in L^1(\mathbb{Q}_p^N)$ and $\phi\in L^2(\mathbb{Q}_p^N)$, $\mathcal{F}(J*\phi)=\widehat{J}\,\widehat{\phi}$.

Conflict of Interests/Competing Interests. I have no conflicts of interest to disclose.

References

[1] Beltrametti E. G., Cassinelli G., Quantum mechanics and p-adic numbers, Found. Phys. 2 (1972), 1–7.

[2] Vladimirov V. S., Volovich I. V., p-adic quantum mechanics, Soviet Phys. Dokl. 33 (1988), no. 9, 669–670.

[3] Vladimirov V. S., Volovich I. V., A vacuum state in p-adic quantum mechanics, Phys. Lett. B 217 (1989), no. 4, 411–415.

[4] Vladimirov V. S., Volovich I. V., p-adic quantum mechanics, Comm. Math. Phys. 123 (1989), no. 4, 659–676.

[5] Volovich I. V., Number theory as the ultimate physical theory, p-Adic Numbers Ultrametric Anal. Appl. 2 (2010), no. 1, 77–87.

[6] Vladimirov V. S., Volovich I. V., Zelenov E. I., p-Adic analysis and mathematical physics, World Scientific, 1994.

[7] Dirac P. A. M., The principles of quantum mechanics. First edition 1930. Fourth Edition, Oxford University Press, 1958.

[8] von Neumann J., Mathematical foundations of quantum mechanics. First Edition 1932. Princeton University Press, Princeton, NJ, 2018.

[9] Zúñiga-Galindo W. A., The p-adic Schrödinger equation and the two-slit experiment in quantum mechanics, Annals of Physics 469 (2024), Paper No. 169747.

[10] Zúñiga-Galindo W. A., p-adic quantum mechanics, the Dirac equation, and the violation of Einstein causality, J. Phys. A 57 (2024), no. 30, Paper No. 305301, 29 pp.

[11] Zúñiga-Galindo W. A., 2-Adic quantum mechanics, continuous-time quantum walks, and the space discreteness, Fortschr. Phys. (2025), e70019. https://doi.org/10.1002/prop.70019.

[12] Zúñiga-Galindo W. A., Mayes N. P., p-Adic quantum mechanics, infinite potential wells, and continuous-time quantum walks, Rev. Math. Phys. https://doi.org/10.1142/S0129055X25500278.

[13] Zúñiga-Galindo W. A., Chacón-Cortés L. F., Continuous-time quantum Markov chains and discretizations of p-adic Schrödinger equations: comparisons and simulations, J. Stat. Mech. Theory Exp. 2026 (2026), no. 9, Paper No. 093401, 55 pp.

[14] Zúñiga-Galindo W. A., p-Adic Dirac equations and the Jackiw–Rebbi model, arXiv:2603.17200.

[15] Zúñiga-Galindo W. A., Quantum mechanics, non-locality, and the space discreteness hypothesis, Annals of Physics 489 (2026), Paper No. 170459.

[16] Zúñiga-Galindo W. A., Wavefunctions localization, and the Wigner's Friend Paradox in a Framework of Discrete-Space Hypothesis, arXiv:2607.00198.

[17] Zúñiga-Galindo W. A., Zambrano-Luna B. A., Indoung C., Pattern formation in quantum hierarchical cellular neural networks, Commun. Nonlinear Sci. Numer. Simul. 163 (2026), part 5, Paper No. 110715.

[18] Bialynicki-Birula I., Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, Phys. Rev. D 49 (1994), no. 12, 6920–6927.
[19] Meyer D. A., From quantum cellular automata to quantum lattice gases, J. Statist. Phys. 85 (1996), no. 5-6, 551–574.
[20] Succi S., Benzi R., Lattice Boltzmann equation for quantum mechanics, Phys. D 69 (1993), no. 3-4, 327–332.
[21] Arrighi P., Nesme V., Forets M., The Dirac equation as a quantum walk: higher dimensions, observational convergence, J. Phys. A 47 (2014), no. 46, 465302, 33 pp.
[22] Childs A. M., Goldstone J., Spatial search and the Dirac equation, Phys. Rev. A 70 (2004), no. 4, 042312.
[23] Bianconi G., The topological Dirac equation of networks and simplicial complexes, J. Phys. Complex. 2 (2021), no. 3, 035022.
[24] Tödtli B., Laner M., Semenov J., Paoli B., Blattner M., Kunegis J., Continuous-time quantum walks on directed bipartite graphs, arXiv:1606.00992.
[25] Wallace P. R., The band theory of graphite, Phys. Rev. 71 (1947), 622–634.
[26] Castro Neto A. H., Guinea F., Peres N. M. R., Novoselov K. S., Geim A. K., The electronic properties of graphene, Rev. Mod. Phys. 81 (2009), 109–162.
[27] Jay G., Debbasch F., Wang J. B., Dirac quantum walks on triangular and honeycomb lattices, Phys. Rev. A 99 (2019), no. 3, 032113, 11 pp.
[28] Arnault P., Debbasch F., Quantum walks and discrete gauge theories, Phys. Rev. A 93 (2016), 052301.
[29] Thaller B., The Dirac equation, Texts Monogr. Phys., Springer-Verlag, Berlin, 1992.
[30] Bjorken J. D., Drell S. D., Relativistic quantum mechanics, McGraw-Hill Book Co., New York-Toronto-London, 1964.
[31] Greiner W., Relativistic quantum mechanics, Springer-Verlag, Berlin, 2000.
[32] Feynman R. P., Hibbs A. R., Quantum Mechanics and Path Integrals, McGraw-Hill, New York, 1965.
[33] Strauch F. W., Relativistic quantum walks, Phys. Rev. A 73 (2006), 054302; erratum Phys. Rev. A 73 (2006), 069908.
[34] Strauch F. W., Relativistic effects and rigorous limits for discrete- and continuous-time quantum walks, J. Math. Phys. 48 (2007), no. 8, 082102, 18 pp.
[35] Kurzyński P., Relativistic effects in quantum walks: Klein's paradox and Zitterbewegung, arXiv:quant-ph/0606171.
[36] Chandrashekar C. M., Banerjee S., Srikanth R., Relationship between quantum walks and relativistic quantum mechanics, Phys. Rev. A 81 (2010), no. 6, 062340, 10 pp.
[37] Gerritsma R., Kirchmair G., Zähringer F., Solano E., Blatt R., Roos C. F., Quantum simulation of the Dirac equation, Nature 463 (2010), 68–71.
[38] Gerritsma R., Lanyon B. P., Kirchmair G., Zähringer F., Hempel C., Casanova J., García-Ripoll J. J., Solano E., Blatt R., Roos C. F., Quantum simulation of the Klein paradox with trapped ions, Phys. Rev. Lett. 106 (2011), no. 6, 060503, 4 pp.
[39] Farhi E., Gutmann S., Quantum computation and decision trees, Phys. Rev. A (3) 58 (1998), no. 2, 915–928.
[40] Mülken O., Blumen A., Continuous-time quantum walks: models for coherent transport on complex networks, Phys. Rep. 502 (2011), no. 2-3, 37–87.
[41] Venegas-Andraca S. E., Quantum walks: a comprehensive review, Quantum Inf. Process. 11 (2012), no. 5, 1015–1106.
[42] Childs A. M., Farhi E., Gutmann S., An example of the difference between quantum and classical random walks, Quantum Information Processing 1 (2002), 35–43. https://doi.org/10.1023/A:1019609420309

[43] Di Molfetta G., Debbasch F., Discrete-time quantum walks: continuous limit and symmetries, J. Math. Phys. 53 (2012), no. 12, 123302, 20 pp.

[44] Di Molfetta G., Brachet M., Debbasch F., Quantum walks as massless Dirac fermions in curved space-time, Phys. Rev. A 88 (2013), no. 4, 042301, 8 pp.

[45] Albeverio S., Khrennikov A. Yu., Shelkovich V. M., Theory of p-adic distributions: linear and nonlinear models, London Mathematical Society Lecture Note Series, 370, Cambridge University Press, 2010.

[46] Reed M., Simon B., Methods of modern mathematical physics. I. Functional analysis, Second edition, Academic Press, New York, 1980.

[47] Pazy A., Semigroups of linear operators and applications to partial differential equations, Applied Mathematical Sciences 44, Springer-Verlag, New York, 1983.

[48] Appell J., Zabrejko P. P., Nonlinear superposition operators, Cambridge Tracts in Mathematics 95, Cambridge University Press, Cambridge, 1990.

[49] Gorini V., Kossakowski A., Sudarshan E. C. G., Completely positive dynamical semigroups of N-level systems, J. Math. Phys. 17 (1976), 821–825.

[50] Lindblad G., On the generators of quantum dynamical semigroups, Comm. Math. Phys. 48 (1976), 119–130.

[51] Breuer H.-P., Petruccione F., The theory of open quantum systems, Oxford University Press, Oxford, 2002.

[52] Zambrano-Luna B. A., Zúñiga-Galindo W. A., p-adic cellular neural networks, J. Nonlinear Math. Phys. 30 (2023), no. 1, 34–70.

[53] Zambrano-Luna B. A., Zúñiga-Galindo W. A., p-adic cellular neural networks: applications to image processing, Phys. D 446 (2023), Paper No. 133668, 11 pp.

[54] Zúñiga-Galindo W. A., Zambrano-Luna B. A., Dibba B., Hierarchical neural networks, p-adic PDEs, and applications to image processing, J. Nonlinear Math. Phys. 31 (2024), 63.

[55] Zúñiga-Galindo W. A., p-Adic analysis: stochastic processes and pseudo-differential equations, De Gruyter, 2025.

[56] Taibleson M. H., Fourier analysis on local fields, Princeton University Press, 1975.

[57] Halmos P., Measure theory, D. Van Nostrand Company Inc., New York, 1950.

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